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FAITH LAUNDRY SHOP MANAGEMENT SYSTEM

A System Documentation for

Faith Laundry Shop Management System

In Partial Fulfillment

of the Requirements for Course

CC 207 Systems Analysis and Design

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Abstract

Faith Laundry Shop, located in Ilaya, Tabuc Suba, Jaro, Iloilo City, is a small-scale laundry service that has been operating since 2022. The business currently relies on manual processes including handwritten logbooks, physical tags, and paper-based receipts for managing daily operations. These methods have led to recurring issues such as time-consuming record keeping, occasional order mix-ups during peak hours, limited order tracking capability, and the absence of automated sales reporting.

This system documentation presents the analysis, design, and development of the Faith Laundry Shop Management System — a web-based application intended to digitize and streamline the shop's core operations. The system supports automated computation of laundry charges based on a per-load pricing structure, real-time order status tracking, digital receipt generation, and automated daily and monthly sales reporting.

The system was developed using a modern technology stack comprising Next.js for the frontend, Spring Boot with Java 21 for the backend, and PostgreSQL for the database. The documentation covers the complete systems development lifecycle, including the case study, requirements specification, system design, implementation, testing, and user documentation.

Keywords: laundry management system, web application, order tracking, automated reporting, Next.js, Spring Boot, PostgreSQL.

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Chapter 1: Planning and Initiation

1.1 Case Study

1.1.1 Introduction

This case study documents the current operational practices of Faith Laundry Shop to identify business problems and opportunities for improvement through an information system.

The study is based on a semi-structured interview and on-site observation with the admin and staff. Findings serve as the foundation for requirements definition, system design, and implementation.

1.1.2 Business Profile

1.1.2.1 Business Overview

Faith Laundry Shop is a small-scale laundry service established in 2022, located in Ilaya, Tabuc Suba Jaro, Iloilo City. The business has operated for approximately three years and caters to local customers requiring regular laundry services.

Operations are managed by the owner with one staff member. Both are directly involved in daily activities.

1.1.2.2 Services Offered

Faith Laundry Shop provides:

- Washing
- Drying
- Folding

Services are offered on a per-load basis, depending on the weight and condition of laundry items.

1.1.3 Current Business Environment

1.1.3.1 Operational Setup

Daily operations are primarily manual. When customers bring laundry, staff measures total weight in kilograms and sorts items by customer preference (e.g., separating colored and white clothing).

The laundry is washed, dried, and folded using washing machines. Completed items are placed in labeled bags with the customer's name and prepared for pickup.

Receipts or claim stubs include:

- Customer name
- Contact number
- Transaction date
- Total payment amount
- Laundry shop name
- Authorized signature

Payment is typically collected upon pickup rather than at drop-off.

1.1.3.2 Pricing Structure

Pricing is computed on a per-load basis:

- One (1) load costs ₱140
- One load covers up to 8 kg

- Laundry exceeding 8 kg is charged as an additional load

Additional charges may apply for:

- Extra fabric conditioner requests
- Extended washing time due to excessive dirt — ₱1 per extra minute of machine use

All pricing is computed manually by staff at the time of transaction.

1.1.3.3 Technology Usage

The business relies on paper-based tools:

- Physical tags
- Logbooks
- Manual notebooks for payments and sales

Records are kept for approximately one month before archiving. No computerized or automated system is used for operations or reporting.

1.1.4 Problem Identification

1.1.4.1 Observed Problems

Based on the interview and observation:

- Manual and Time-Consuming Record Keeping — Recording orders, payments, and sales manually requires significant effort from Admin and staff.
- Occasional Order Mix-Ups — During peak hours or with multiple rush orders, clothes may be mixed, increasing error risk and customer dissatisfaction.

- Limited Order Tracking — Laundry status depends on logbooks and physical inspection, leading to delays or inaccuracies.
- Lack of Automated Reporting — Income tracking is manual; the admin lacks automated daily, monthly, or yearly reports.
- Operational Dependency on Machinery — Washing machine malfunctions cause stress and financial impact. Better tracking and visibility can help manage operations during downtime.

1.1.4.2 Root Causes

- Heavy reliance on manual processes
- Lack of system automation
- Dependence on human memory for order tracking

These factors limit efficiency, increase error likelihood, and restrict access to timely business information.

1.1.5 Business Impact

The identified problems affect operational efficiency and reliability. Manual record keeping slows service and increases inaccuracy risk. Order mix-ups can lead to customer complaints and loss of trust.

The absence of automated reporting makes it difficult for the admin/owner to evaluate performance, compare income across periods, and make informed decisions.

1.1.6 Opportunity for System Improvement

Findings indicate a strong opportunity to improve operations through a Laundry Shop Management System that supports:

- Digital recording of laundry orders
- Centralized transaction history
- Automated computation of laundry charges
- SMS or digital notifications for customers
- Order tracking using a unique reference number

Providing staff and customers with a reliable way to check order status reduces reliance on verbal inquiries and manual logbooks, thereby improving service efficiency and accuracy.

1.1.7 Project Justification

Developing an information system is justified by the need to streamline operations, reduce errors, and support better decision-making.

The business owner has expressed willingness to adopt a simple computer-based system. A system-based solution will improve efficiency, enhance customer service, and support sustainable growth.

1.1.8 Conclusion

This case study examined current operations of Faith Laundry Shop and identified challenges related to manual processes, limited tracking, and lack of automated reporting.

The findings demonstrate a clear need for a system-based solution and provide the foundation for requirements definition, system design, and implementation.

1.2 System Service Request

Requested by: Rowelen Planco Cornelio

Company/Client: Faith Laundry Shop

Title/Position: Admin

Location: Sitio Ilaya, Tabuc Suba, Jaro, Iloilo City

Phone: 09291554954

Email: corneliorowelen@yahoo.com

Type of Request

☑ New System
☐ System Enhancement
☐ System Error Correction
☐ Other:

Urgency Level

☐ Immediate: Operations are impaired or opportunity lost
☐ Problems exist, but can be worked around
☑ Business losses can be tolerated until new system installed

1.2.1 Problem Statement

1.2.1.1 Background

Faith Laundry Shop has experienced steady operations over the past three years, handling multiple customer transactions daily. As transaction volume increases, the business has become more dependent on manual processes to manage laundry orders, payments, and records.

1.2.1.2 Current System Overview

At present, all operational activities are handled manually. Laundry orders are tracked using physical tags, logbooks, and staff recall. Payments and daily sales are recorded in notebooks, and receipts are issued manually. Business records are paper-based and are kept only for a limited period before being archived. No automated system is currently in use.

1.2.1.3 Problems Identified

The current system presents the following business problems:

- Manual and time-consuming record keeping, increasing workload and reducing efficiency.
- Occasional order mix-ups, particularly during peak hours or rush orders.
- Limited ability to track laundry order status, relying on logbooks and physical inspection.
- Lack of automated reports, making it difficult to monitor daily, monthly, and yearly income.
- Operational dependency on washing machines, where malfunctions lead to downtime and financial loss.

1.2.1.4 Business Impact

These problems negatively affect operational efficiency, increase the likelihood of human error, and limit the owner's ability to assess business performance accurately. The absence of timely and reliable information restricts planning and decision-making, while order mix-ups and delays may impact customer satisfaction and trust.

1.2.2 Service Request Description

Faith Laundry Shop formally requests the analysis, design, and development of a Laundry Shop Management System that will support the digital recording and tracking of laundry orders and transactions. The proposed system is intended to replace manual processes and provide centralized access to accurate operational data.

The system will support existing business practices, including per-load pricing computation, order status tracking, and transaction recording, while allowing flexibility for future adjustments to business rules.

1.2.3 Expected Business Benefits

The implementation of the proposed system is expected to deliver the following benefits:

- Improved efficiency through automation of order tracking and record keeping
- Reduced risk of errors and order mix-ups
- Faster and more accurate access to sales and transaction data
- Enhanced monitoring of business performance through automated reports
- Improved customer service and communication

1.2.4 Approval

☑ Request approved
☐ Recommend Revision
☐ Suggest User Development
☐ Reject for reason:

Assigned to:	HIMÓTECH Development Team
Approval Date:	2026-02-08

1.3 Project Scope and Limitations

1.3.1 Project Scope

The Faith Laundry Shop Management System is designed to automate and streamline the core daily operations of the laundry shop. The system will cover the following key areas:

- Transaction Management: Recording of customer drop-offs, weight in kilograms, calculation of total loads, and automatic computation of fees (including extra charges like extended wash time or fabric conditioner).
- Customer Management: Maintaining a basic database of customer names and contact numbers for transaction tracking and future SMS notifications.
- Order Tracking: Monitoring the status of laundry through the stages: Received, Washing, Drying, Folding, Ready for Pickup, and Released.
- Payment Processing: Recording payments (paid upon pickup) and issuing digital or printable receipts.
- Reporting & Analytics: Generating daily, weekly, and monthly sales reports for the owner/admin to track income and performance.

1.3.2 Project Limitations

To ensure a successful deployment within the allotted timeframe, the system has the following limitations:

- No Online Payment Gateway: The system records payments made via Cash, GCash, or Bank Transfer for bookkeeping purposes only. It does not integrate directly with external payment gateway APIs (e.g., GCash API, Maya API) to process transactions electronically.
- No Customer Account System: Customers do not have user accounts or login credentials. However, a public order tracking portal allows customers to check their laundry status using the order reference number without authentication.

- No Inventory Management for Supplies: The system will not track the volume of detergent or fabric conditioner remaining in the shop.
- No Hardware Integration: The system will not automatically pull data from physical weighing scales or washing machines; data must be manually entered by the staff.

Chapter 2: Project Planning and Management

2.1 Stakeholder Analysis

2.1.1 Purpose of the Stakeholder Analysis

The purpose of this stakeholder analysis is to identify individuals and groups who have an interest in or influence over the Laundry Shop Management System. This analysis informs the project manager and sponsor of stakeholder expectations, potential risks, and the actions required to ensure stakeholder engagement prior to and during detailed project planning.

2.1.2 Stakeholder Analysis Matrix

Table 1. Stakeholder Analysis Matrix

Stakeholder	Their interest or requirement for the project	What the project needs from them	Perceived attitudes and/or risks	Actions to take
Owner	• Improved operational efficiency • Reduced manual record keeping • Accurate daily, monthly, and yearly sales reports • Better control over business operations	• Approval of project scope and requirements • Participation in interviews and review meetings • Validation of system outputs and reports	• Concern about system complexity and Data Security • High interest and strong support for the system • Risk of limited availability due to daily operational duties	• Schedule regular review and approval meetings • Validate requirements and reports early • Involve Admin in major decision points • Keep system simple and practical • Present clear prototypes for feedback
Staff	• Easier	• Accurate	• Generally	• Conduct

	tracking of laundry orders Reduced errors during order handling • Simple and easy-to-use system interface	entry of laundry orders and status updates • Feedback on workflow and usability during analysis and prototyping • Participation in observation and validation	supportive of the project • Possible resistance to changes in routine or unfamiliar technology • Fear of making mistakes using a computer	process walkthroughs with staff • Provide basic system orientation and guidance • Gather usability feedback during prototype reviews
Customers	• Accurate handling of laundry items • Timely completion of laundry services • Clear communication regarding order status and pickup	• Acceptance of system-generated receipts or notifications • Indirect feedback on service quality and system effectiveness	• Neutral to positive attitude toward system changes • Risk of dissatisfaction if errors or delays continue during transition	• Ensure system improves order accuracy • Minimize service disruption during implementation • Communicate service process clearly to customers
System Development Team	• Clear and complete business requirements • Timely feedback from stakeholders • Access to users for	• Proper analysis, design, and documentation of the system • Delivery of the system according to agreed scope	• Risk of requirement changes or delayed feedback • Risk of schedule impact due to stakeholder	• Establish clear communication channels • Document and confirm requirements formally • Define review

	clarification and validation	and schedule	availability	milestones and approval checkpoints

2.2 Roles and Responsibilities

2.2.1 Purpose

This Roles and Responsibilities Matrix define governance, management, development, deployment, and operational roles for the Laundry Shop Management System project. It ensures accountability, alignment with SDLC principles (SAD 10th Edition), and clear Adminship of deliverables across all phases of the project lifecycle.

2.2.2 Project Roles and Responsibilities Matrix

Table 2. Roles and Responsibilities Matrix

Role Category	Role	Responsibilities	Responsible Individual / Team
Governance	Project Sponsor (Business Owner)	• Provides overall accountability for the project • Approves scope, requirements, and major deliverables • Ensures alignment with business objectives Authorizes transition between SDLC phases	Rowelen Planco Cornelio (Owner)
Management	Project Manager / System Analyst	• Plans and manages project activities • Coordinates	Christian De la Cruz

		planning, analysis, design, and implementation • Manages schedule and scope • Communicates progress to sponsor • Ensures SDLC compliance	
	Risk & Quality Coordinator	• Monitors project risks • Ensures documentation consistency • Verifies alignment between requirements and deliverables • Reviews outputs for quality control	Alyanna Bianca Serra
Development	Business Analyst	• Analyzes current business processes Documents functional and non-functional requirements • Validates requirements with stakeholders • Supports process and data modeling	Eliza May Calisa
	System Architect	• Defines system architecture (web-based structure) • Determines technology stack	Mark Alvin Cadangin

		• Ensures scalability and maintainability • Oversees integration between components	
	UI/UX Designer	• Designs system interface and navigation • Applies usability principles • Implements input validation and error handling standards	Ellen Mae Tacleon
	Software Developer(s)	• Implements system modules (order tracking, pricing computation, reporting) • Develops database integration • Ensures system functionality meets requirements	Mark Alvin Cadangin Luisa Rose Brillantes Christian De la Cruz Alyanna Bianca Serra Eliza May Calisa Ellen Mae Tacleon
	Tester / Quality Assurance	• Develops test cases and test plans • Conducts functional and usability testing • Documents and tracks defects • Verifies system against requirements	Luisa Rose Brillantes
Deployment	Deployment & Configuration Lead	• Prepares system deployment	Christian De la Cruz

		environment • Configures application and database • Ensures system installation readiness	
	Training & Documentation Lead	• Prepares user manuals and documentation • Conducts user training sessions • Documents system procedures	Eliza May Calisa
Operations	System Administrator	• Manages user access and system configuration • Monitors system usage • Ensures operational stability	Rowelen Planco Cornelio (Admin)
	Data & Backup Manager	• Implements data backup procedures Maintains data integrity • Oversees recovery procedures	Mark Alvin Cadangin Luisa Rose Brillantes
	User Support	• Provides assistance to system users • Logs issues and feedback • Escalates technical problems when needed	Christian De la Cruz Alyanna Bianca Serra Eliza May Calisa Ellen Mae Tacleon

2.3 Work Breakdown Structure

2.3.1 Overview

The Work Breakdown Structure hierarchically decomposes the Faith Laundry Shop Management System into manageable phases and deliverables. This ensures that all project requirements are met systematically.

2.3.2 Work Breakdown Structure (WBS)

2.3.2.1 Phase 1: Planning and Initiation

- Initial Client Interview & Case Study
- Identify Business Problems
- System Service Request (SSR)
- Project Charter & Scope Definition
- Stakeholder Analysis & Resource Alignment

2.3.2.2 Phase 2: Requirements Analysis

- Gather Functional Requirements
- Gather Non-Functional Requirements
- Define User Stories & Business Rules
- As-Is Process Modeling
- To-Be Process Modeling
- Data Flow & Functional Decomposition Modeling
- Requirements Validation Sign-off

2.3.2.3 Phase 3: System Design

- Entity Relationship Diagram (ERD) & Database Schema Design
- UI/UX Wireframing & Prototyping
- API Architecture Design (OpenAPI)
- Technology Stack Selection

2.3.2.4 Phase 4: Development

- Environment Setup (Backend & Database)
- Backend Implementation (Authentication, Endpoints)
- Frontend Implementation (Dashboard, Order Tracking, Computations)
- System Integration

2.3.2.5 Phase 5: Testing

- Unit Testing
- 5.2 System Integration Testing
- 5.3 Usability Testing (UI/UX Validation)
- 5.4 User Acceptance Testing (UAT)

2.3.2.6 Phase 6: Implementation & Deployment

- Server & Database Provisioning
- Application Deployment
- Post-Deployment Validation
- System Go-Live

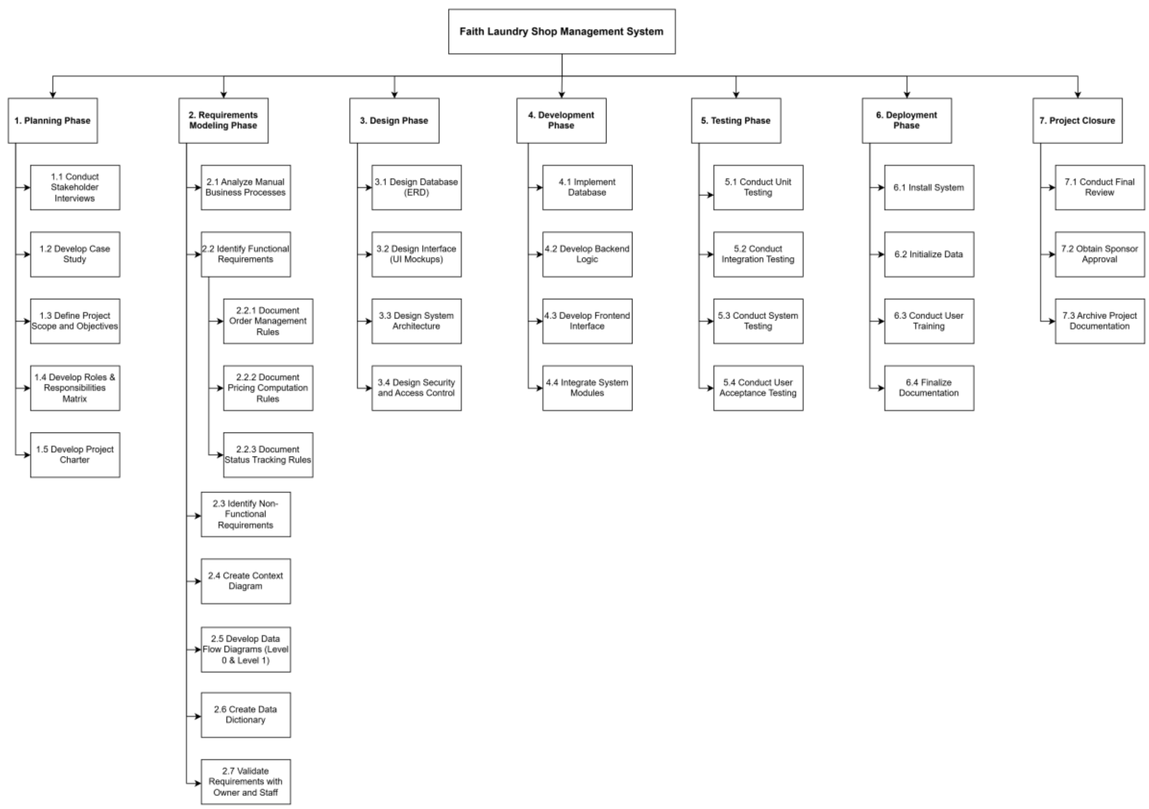
2.3.2.7 Phase 7: Handover & Closure

- User Manuals & Training
- Handover Session Checklist & Sign-off
- Project Documentation Archive

2.3.3 WBS Diagram

Below is the visual representation of the Work Breakdown Structure.

Figure 1. Work Breakdown Structure Diagram



2.4 Project Schedule

2.4.1 Purpose

This outlines the project schedule, task dependencies, and critical path for the development of the Faith Laundry Shop Management System. It utilizes a Gantt Chart for chronological visualization and a PERT/CPM matrix to identify the sequence of crucial activities required to complete the project within the 12-week academic term.

2.4.2 Activity Mapping

The following table defines the major project activities used in the scheduling diagrams and maps them directly to the specific deliverables outlined in the Work Breakdown Structure (WBS).

Table 3. Activity Mapping

Activity ID	**Activity Description**	**WBS Reference**
A	Initial Planning & Case Study	1.1, 1.2
B	Charter & Objectives Definition	1.3, 1.4, 1.5
C	Requirements Identification	2.1, 2.2, 2.3
D	Data/Process Modeling & Validation	2.4, 2.5, 2.6, 2.7
E	Database & Architecture Design	3.1, 3.3, 3.4
F	Interface Design (UI Mockups)	3.2
G	Database & Backend Development	4.1, 4.2

H	Frontend Development	4.3
I	Integration & Testing	4.4, 5.1, 5.2, 5.3, 5.4
J	Deployment & Closure	6.1, 6.2, 6.3, 6.4, 7.1, 7.2, 7.3

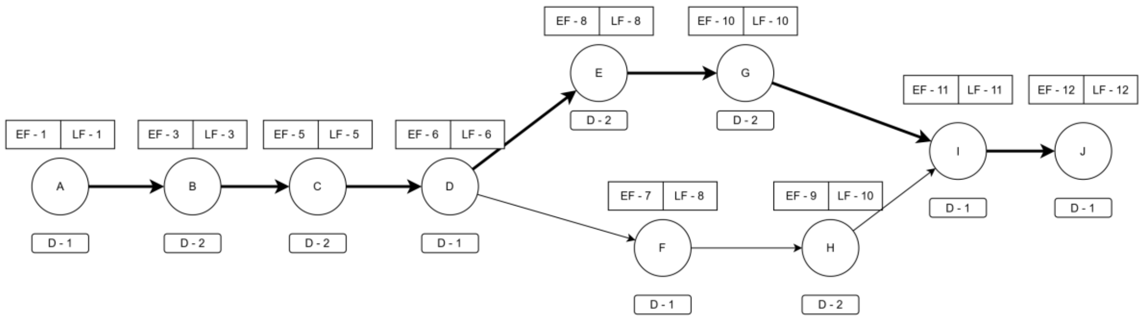
2.4.3 PERT/CPM Analysis

The Expected Duration is calculated in Weeks. The mathematical logic below ensures milestones are met: Planning (Week 3), Requirements (Week 6), Design (Week 8), Development (Week 10), and Deployment (Week 12).

Table 4. PERT/CPM

Activity	**Preceding Event**	**Expected Duration**	**EF**	**LF**	**Slack**	**Critical Path?**
A	-	1	1	1	0	Y
B	A	2	3	3	0	Y
C	B	2	5	5	0	Y
D	C	1	6	6	0	Y
E	D	2	8	8	0	Y
F	D	1	7	8	1	N
G	E	2	10	10	0	Y
H	F	2	9	10	1	N
I	G, H	1	11	11	0	Y
J	I	1	12	12	0	Y

Figure 2. PERT/CPM Network Diagram



2.4.4 Gantt Chart

The timeline below illustrates the planned progression of activities across the 12-week schedule.

Table 5. Gantt Chart

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
A												
B												
C												
D												
E												
F												
G												
H												
I												
J												

2.5 Risk Management Plan

2.5.1 Purpose

This identifies potential risks that could impact the successful development, deployment, and adoption of the Faith Laundry Shop Management System. It evaluates the probability and impact of each risk and establishes actionable mitigation strategies to minimize disruptions to the project schedule and the daily operations of the business.

2.5.2 Risk Identification & Mitigation Matrix

Table 6. Risk Identification & Mitigation Matrix

Risk ID	Risk Category	Risk Description	Probability	Impact	Mitigation Strategy	Risk Admin
R01	User Adoption	Staff resistance to unfamiliar technology or fear of making computer errors.	Medium	High	Conduct comprehensive process walkthroughs and provide basic system orientation. Ensure the UI design is simple, practical, and heavily validated during prototyping.	Training Lead
R02	Schedule	Project delays due to the limited availability of the Business Admin (Sponsor) because of daily operational duties.	High	Medium	Schedule regular, brief review meetings aligned with the owner's availability. Present clear prototypes for quick feedback and establish firm approval checkpoints.	Project Manager
R03	Technical / Security	Admin's concern regarding	Low	High	Implement standard database	System Architect

		data security and system complexity.			security measures, secure role-based access control (Admin vs. Staff), and regular data backup protocols.	
R04	Scope	Requirement changes or delayed feedback impacting the 12-week academic SDLC timeline.	Medium	High	Document and confirm all requirements formally via sign-offs. Strictly control scope creep (e.g., firmly exclude mobile apps or SMS gateways as defined in the charter).	Business Analyst
R05	Operational Transition	Customer dissatisfaction if service delays or errors occur during the transition from the manual logbook to the digital system.	Medium	Medium	Minimize service disruption by launching the system in a pilot form. Clearly communicate the new order reference tracking process to customers during the transition.	Deployment Lead

2.5.3 Contingency Plan

In the event of critical system failure or prolonged power outage during the pilot deployment, the shop will temporarily revert to the existing paper-based logbook and physical tagging system to ensure zero interruption to customer service. All manual records will be retroactively entered into the system by the Data Manager once operations normalize.

Chapter 3: Systems Analysis and Requirements

3.1 Process Matrix

3.1.1 Purpose of the Process Matrix

This Process Matrix the current (As-Is) and proposed (To-Be) business processes of Faith Laundry Shop. It identifies manual activities, inefficiencies, and error-prone steps and defines how the proposed Laundry Shop Management System will support existing business practices without altering the core service model. This matrix serves as the primary basis for functional requirements, system modeling, and design.

3.1.2 Process Matrix

Table 7. Process Matrix

Process ID	Business Process	Primary Actor(s)	Inputs	Process Description (As-Is)	Outputs	System Support (To-Be)
P-01	Receive Laundry Order	Admin / Staff	• Customer laundry • Customer name • Contact number	• Receive laundry from customer. • Weigh laundry in kilograms. • Sort items based on customer preferences (e.g., white vs colored).	• Accepted laundry order	• Record customer and order details digitally. • Generate unique order reference number. • Set initial order status to Received.
P-02	Compute	Admin /	• Laundry	• Compute	• Total amount	• Automatically

	Laundry Charges	Staff	weight • Current pricing rules	charges manually on a per-load basis. • One load covers up to 8 kg at standard rate. • Add extra charges for extended washing time when applicable.	due	compute charges based on configurable pricing rules. Allow adjustments for special cases.
P-03	Issue Receipt / Claim Stub	Admin / Staff	• Order details • Computed charges	• Write receipt manually. Provide customer with handwritten claim stub.	• Receipt / claim stub	• Generate system-based receipt. • Display order reference number clearly. • Store transaction details.
P-04	Process Laundry	Staff	• Laundry order	• Wash, dry, fold, and iron laundry items. • Place completed laundry in labeled bags.	• Processed laundry	• Update order status progressively through Washing, Drying, Folding, and Ready for Pickup.

P-05	Track Laundry Status	Admin / Staff	• Logbook • Physical tags	• Check logbook or physical tags. • Rely on memory to identify order progress.	• Order status information	• View real-time order status using order reference number.
P-06	Record Payment	Admin / Staff	• Customer payment	• Accept payment, typically upon pickup. • Record payment manually in notebook.	• Payment record	• Record payment digitally. • Link payment to corresponding order.
P-07	Release Laundry	Staff	• Claim stub Completed laundry	• Verify customer claim stub. • Check laundry condition and quantity. • Release laundry to customer.	• Released laundry	• Validate order using order reference number. • Update order status to Released.
P-08	Maintain Records	Admin	• Paper-based records	• Store receipts and notebooks. • Archive	• Archived records	• Maintain centralized digital records with extended retention.

				records after approx. one month.		
P-09	Generate Reports	Admin	• Transaction records	• Manually compute income totals.	• Daily/Monthly income	• Generate automated daily, monthly, and yearly sales reports.
P-10	Provide Order Status Information	Admin / Staff	• Order reference number	• Respond to customer inquiries verbally.	• Customer updates	• Allow staff and customers to check order status using the order reference number.

3.2 Business Rules

This document outlines the core business logic and computational rules that the Faith Laundry Shop Management System must enforce. Each rule is categorized by domain and assigned a unique identifier for traceability to the functional requirements and test cases.

3.2.1 Pricing and Load Rules (BR-PR)

- BR-PR-01 (Base Load Pricing): One (1) standard load is defined as a maximum of eight (8) kilograms. The base price for one (1) standard load of "Standard Wash" is ₱140.00.
- BR-PR-02 (Additional Load for Excess Weight): If a customer's laundry exceeds the per-load weight limit (8 kg for Standard Wash), the total number of loads is

computed as ceil(weightkg / kglimitperload). Each additional load is charged at the same base price per load (e.g., 9 kg = 2 loads = ₱280.00).

- BR-PR-03 (Extra Washing Time Charge): Washing time beyond the included 45 minutes per load is charged at ₱1.00 per extra minute. If no extra minutes are recorded, this charge is zero.
- BR-PR-04 (Optional Add-ons): Additional requests (e.g., fabric conditioner) must add standard predefined amounts to the total bill. Add-on pricing is itemized as separate line items on the order.
- BR-PR-05 (Admin Controls Service Rates): The Admin must be able to update service rates (base price per load, kg limit per load, price per extra minute) without code changes. Rate changes apply only to new orders; existing orders retain the snapshot pricing captured at creation time.

3.2.2 Order Lifecycle Rules (BR-OL)

- BR-OL-01 (Unique Reference Number): Every order must have a unique reference number in the format LDR-XXXXXXXX-XXXX used for tracking and customer claim stubs. The system generates this automatically upon order creation.
- BR-OL-02 (Initial Order Status): A newly created order must start with status RECEIVED. No other initial status is allowed.

- BR-OL-03 (Allowed Order Statuses): Order status must be one of the following: RECEIVED, WASHING, DRYING, FOLDING, READYFORPICKUP, RELEASED, or CANCELLED.
- BR-OL-04 (Status Transition Sequence): Orders should follow a logical sequence of stages: RECEIVED → WASHING → DRYING → FOLDING → READYFORPICKUP → RELEASED. Any non-terminal status may transition to CANCELLED. The CANCELLED and RELEASED statuses are terminal — no further transitions are allowed.
- BR-OL-05 (Release Preconditions): An order can only be released (status changed to RELEASED) if two conditions are met: (1) the current status is READYFORPICKUP, and (2) the payment status is PAID. The system must reject release attempts for unpaid orders.
- BR-OL-06 (Order Edit Restrictions): Staff or Admin may edit an order's extra minutes and add-ons only when the order is unpaid and not released. Weight, base amount, and total loads are immutable after creation.

3.2.3 Payment Rules (BR-PAY)

- BR-PAY-01 (Payment Timing): Payment is typically collected upon pickup, not at drop-off. This is a process guideline, not a strict system constraint.
- BR-PAY-02 (Payment Linked to Order): Each payment must be associated with exactly one order. This is enforced by a database foreign key constraint.

- BR-PAY-03 (Payment Amount Validation): The recorded payment amount must exactly match the order grand total. Partial payments, overpayments, and refunds are not supported in the current version.
- BR-PAY-04 (Payment Status): An order's payment status is either PAID or UNPAID. The status is derived from the presence of a completed payment record.
- BR-PAY-05 (Payment Method): Each payment must record the payment method used: Cash, GCash, or Bank Transfer. This is for record-keeping only; the system does not integrate with external payment gateways.
- BR-PAY-06 (Automatic Reversal on Cancellation): When a paid order is cancelled, the associated payment status must automatically transition to VOIDED to ensure financial reports accurately reflect net revenue.

3.2.4 Records and Retention Rules (BR-REC***)***

- BR-REC-01 (Core Data Requirements): The system must store, at minimum: customer name and contact number, complete laundry order details (service type, weight, loads, pricing breakdown), payment records, and all data necessary to derive daily sales totals.

3.2.5 Notification Rules (BR-NOTIF)

- BR-NOTIF-01 (Customer Ready Notification): The system should generate a notification (via SMS through Semaphore.co) when the order status changes to READYFORPICKUP. This notification includes the order reference number.

- BR-NOTIF-02 (Tracking by Reference Number): Customers must be able to track their laundry status using the order reference number through a public tracking portal. No sensitive or internal data shall be exposed through this portal.

3.3 User Stories

This document describes the required system functionality from the perspective of the end-users. Each user story follows the standard format: "As a [role], I want [capability] so that [benefit]." Acceptance criteria define the conditions that must be satisfied for a story to be considered complete.

3.3.1 Epic 1: Order Intake and Management

3.3.1.1 US-01 — Record Laundry Order

As a staff or admin, I want to record a laundry order with customer and laundry details so that the system can track the order from drop-off to pick up.

- Acceptance Criteria:
- Customer name and contact number are required.
- Laundry weight (kg) is required.
- System generates a unique reference number in the format LDR-XXXXXXXX-XXXX.
- Initial order status is set to Received.

3.3.1.2 US-02 — Automatically Compute Laundry Price

As a staff or admin, I want the system to compute the laundry price automatically so that pricing errors and manual calculations are avoided.

Acceptance Criteria:

- One (1) load covers up to 8 kg; total loads are computed as ceil(weightkg / kglimit).
- Base pricing is determined by the selected service rate (e.g., Standard Wash at ₱140/load).
- Each load includes up to 45 minutes of washing time in the base price.
- Extra minutes beyond 45 per load are charged at the per-extra-minute rate (e.g., ₱1.00/min for Standard Wash).
- Staff records extra minutes before the system computes the final price.
- Computed total (base amount + extra minutes + add-ons) is displayed before saving.

3.3.1.3 US-03 — Update Laundry Order Status

As a staff or admin, I want to update the laundry order status so that the current processing stage is accurately reflected.

Acceptance Criteria:

- Allowed statuses: Received, Washing, Drying, Folding, Ready for Pickup, Released, Cancelled.
- Status changes are recorded with a timestamp and the user who performed the change.
- Only existing orders can have their status updated.
- Orders follow a logical status progression; backward transitions are not allowed.

3.3.2 Epic 2: Order Tracking and Release

3.3.2.1 US-04 — Track Laundry Order by Reference Number

As a customer, I want to track my laundry order using a reference number so that I can check the status without asking the staff.

Acceptance Criteria:

- Customer can enter an order reference number on the public tracking page.
- System displays current order status, order date, service summary, and timestamped status history.
- Invalid reference numbers show a clear error message.
- No sensitive or internal data (e.g., pricing, staff details) is exposed.

3.3.2.2 US-05 — Verify and Release Laundry

As a staff or admin, I want to verify laundry details and record payment before releasing the order so that incorrect items are not given to customers and revenue is collected.

Acceptance Criteria:

- Staff can view order and customer details before release.
- Order must be in Ready for Pickup status before release.
- Payment must be recorded (Paid) before release — release is not allowed for unpaid orders.
- Order status is updated to Released after successful verification and payment.

3.3.3 Epic 3: Payments and Transactions

3.3.3.1 US-06 — Record Payment for Laundry Order

As a staff or admin, I want to record customer payments so that payment history is properly tracked.

Acceptance Criteria:

- Payment is linked to exactly one order.
- Payment amount must exactly match the order grand total (full payments only in the current version).
- Payment method (Cash, GCash, or Bank Transfer) is recorded for each payment.
- Payment date is recorded automatically.
- Order payment status is updated to Paid upon successful recording.

3.3.3.2 US-07 — View Payment History

As the admin, I want to view payment records so that I can review transaction history.

Acceptance Criteria:

- Payments are displayed in a paginated, searchable table.
- Each payment shows order reference number, customer name, amount, payment method, receiving staff, and date.
- Payment records are accessible to both Admin and Staff roles.

3.3.4 Epic 4: Records and Reporting

3.3.4.1 US-08 — View Daily Sales Report

As the admin, I want to view daily sales automatically so that I do not need to compute income manually.

Acceptance Criteria:

- System shows total daily revenue.
- The number of paid orders is displayed.
- Average order value is computed.
- Data is based on recorded payments only; voided payments are excluded.

3.3.4.2 US-09 — View Monthly Income Reports

As the admin, I want to view monthly income reports so that I can monitor business performance over time.

Acceptance Criteria:

- Reports can be filtered by month and year.
- Total income is computed automatically.
- Revenue charts and order volume trends are displayed visually.
- Only paid and non-voided orders are included in revenue calculations.

3.3.5 Epic 5: Customer Communication

3.3.5.1 US-10 — Notify Customer When Laundry Is Ready

As a customer, I want to receive a notification when my laundry is ready so that I know when to pick it up.

Acceptance Criteria:

- Notification is triggered when order status becomes Ready for Pickup.
- Notification includes the order reference number.
- Notification channel is SMS via the Semaphore.co API.
- Client alert records are logged in the system for audit purposes.

3.3.6 Epic 6: User Access and Control

3.3.6.1 US-11 — User Login and Role-Based Access

As a system user, I want to log in based on my role so that I can access appropriate system features.

Acceptance Criteria:

- User roles: Admin and Staff.
- Admin: full access to all modules including reports, user management, service rates, and audit logs.
- Staff: manage orders, customers, and payments; no access to reports, user management, or service rate configuration.
- Authentication uses JWT tokens with session-based expiration.

3.4 Use Cases

3.4.1 Use Case Diagram Description

Actors:

- Staff: Handles day-to-day operations including creating orders, updating status, recording payments, and managing customers.

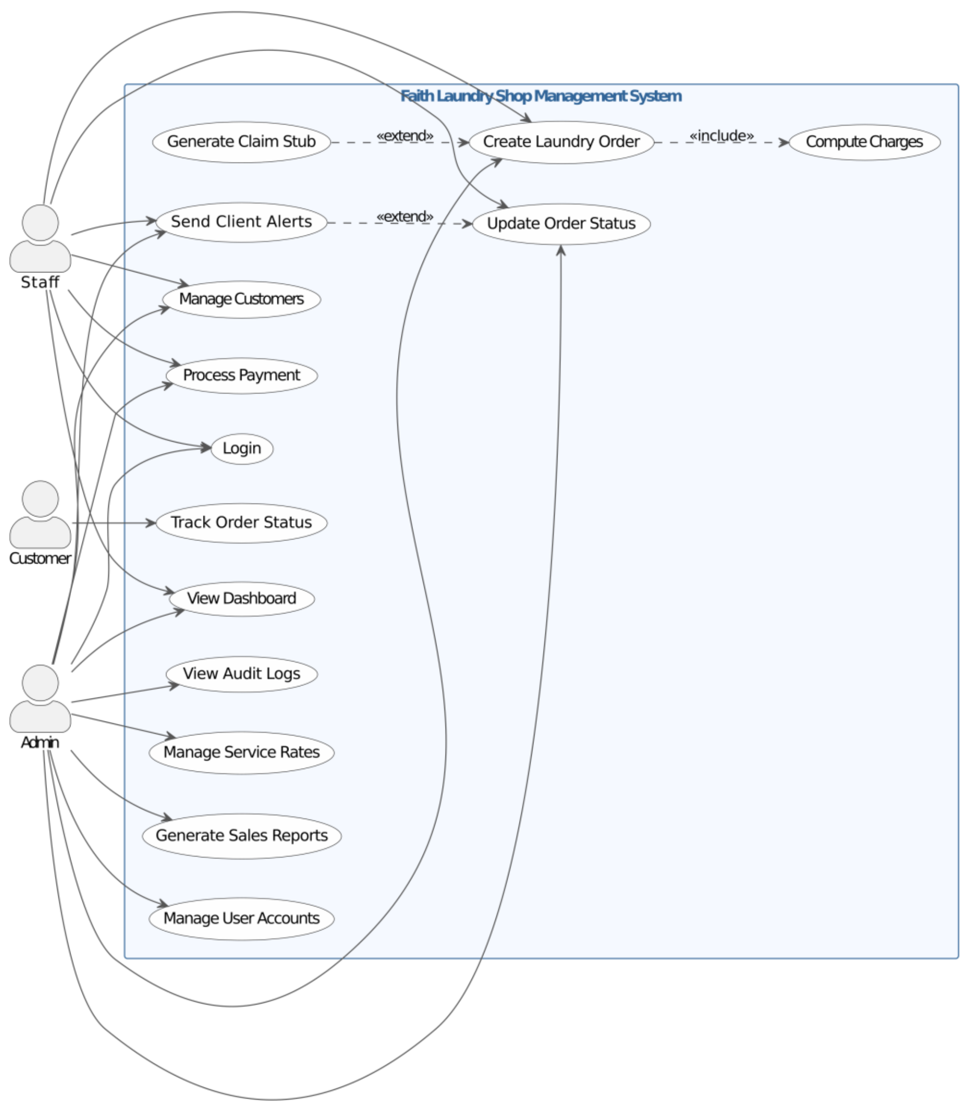
- Admin: Oversees the entire shop, views reports, manages system settings (service rates, user accounts), and reviews audit logs.
- Customer (External): Tracks order status via the public tracking portal. Does not have system credentials.

Core Use Cases:

- Manage Customers (Staff, Admin)
- Manage Laundry Orders (Staff, Admin)
- Process Payments (Staff, Admin)
- Track Order Status (Customer — public)
- Generate Sales Reports (Admin only)
- Manage Service Rates (Admin only)
- Manage Users (Admin only)
- View Audit Logs (Admin only)
- Send Customer Notifications (System — automated)

3.4.2 Use Case Diagram

Figure 3. Use Case Diagram



3.5 Functional Requirements Checklist

3.5.1 Business Process Requirements

Table 8. Business Process Requirements

ID	Requirement	Priority
1.1	The system shall allow staff to create a laundry order by inputting customer details, laundry weight, and service type (e.g., regular, rush, ironing, blankets).	☑ Mandatory ☐ Desirable
1.2	The system shall automatically compute the total price based on the weight, service type, and any additional fees, replacing manual computation.	☑ Mandatory ☐ Desirable
1.3	The system shall allow staff to update and track the exact status of a laundry order (e.g., Received, Washing, Drying, Folding, Ready for Pickup, Released).	☑ Mandatory ☐ Desirable
1.4	The system shall link the payment status (Paid, Unpaid) directly to the corresponding laundry order to prevent confusion during release.	☑ Mandatory ☐ Desirable
1.5	The system shall generate a digital or printable claim stub containing the customer's name, weight, total price, and expected pickup date.	☑ Mandatory ☐ Desirable

3.5.2 User Interface Requirements

Table 9. User Interface Requirements

ID	Requirement	Priority

2.1	The system shall display a dashboard providing the admin and staff with real-time visibility of pending orders, orders ready for pickup, and daily transactions.	☑ Mandatory ☐ Desirable
2.2	The system shall provide visual alerts or highlights for "Rush Orders" to prioritize processing and prevent delays.	☑ Mandatory ☐ Desirable

3.5.3 Data Requirements

Table 10. Data Requirements

ID	Requirement	Priority
3.1	The system shall automatically assign and store a unique order reference number for every laundry order to prevent the misidentification and mixing of clothes.	☑ Mandatory ☐ Desirable
3.2	The system shall securely store a centralized database of all customer transactions, order histories, and payment records.	☑ Mandatory ☐ Desirable

3.5.4 Category: Security Requirements

Table 11. Security Requirements

ID	Requirement	Priority
4.1	The system shall require users to authenticate with a username and password before accessing the system.	☑ Mandatory ☐ Desirable
4.2	The system shall restrict the deletion or modification of completed transactions and payment records to the admin/manager account only.	☑ Mandatory ☐ Desirable

3.5.5 Category: Reporting and Analytics Requirements

Table 12. Reporting and Analytics Requirements

ID	Requirement	Priority
5.1	The system shall automatically generate daily and monthly sales reports so the admin can view total income without manual computation.	☑ Mandatory ☐ Desirable
5.2	The system shall produce a report of historical transactions to allow the admin to analyze business trends over time.	☑ Mandatory ☐ Desirable

3.6 Functional Requirements Matrix

3.6.1 Laundry Shop Management

Table 13. Laundry Shop Management

Process ID	Must?	The System Must or Should
1	✓	Allow the user to create a new laundry order by inputting the customer's name, laundry weight, and specific service type (e.g., regular, rush, blankets).
2	✓	Automatically compute the total transaction price based on the recorded weight, selected service type, and any additional fees, eliminating manual calculation.
3	✓	Allow users to update and track the real-time status of each laundry order (e.g., Received, Washing, Drying, Folding, Ready for Pickup, Released).
4	✓	Link payment records directly to the specific laundry order to clearly indicate whether an order is "Paid" or "Unpaid" before staff release the items to the customer.
5	✓	Reproduce (display and print) a transaction claim stub containing the customer's name, order weight, total price, and expected pickup date.
6	✓	Reproduce (display and print) a dashboard or list showing all current orders, specifically filtering those that are pending, processing, or ready for pickup.
7	✓	Highlight or provide visual alerts for "Rush Orders" to help staff prioritize processing and prevent delays.
8	✓	Reproduce (display and print) daily and monthly sales reports summarizing total income and the number of transactions processed.
9	✓	Automatically assign a unique transaction ID to every laundry order to prevent the mixing or misidentification of customers' clothes.
10	✓	Restrict the ability to modify or delete completed transactions and payment records to the admin/manager account only.

3.7 Non-Functional Requirements

3.7.1 Security

Table 14. Security

ID	Requirement	Priority
NFR-S1	The system SHALL authenticate users (Admin, Staff) via username and password.	☑ Mandatory ☐ Desirable
NFR-S2	The system SHALL use JWT (or equivalent) for session management; credentials SHALL NOT be stored in client-side code.	☑ Mandatory ☐ Desirable
NFR-S3	The system SHALL enforce role-based access: Admin (reports, rates, all operations); Staff (orders, status, payments, customers; no reports).	☑ Mandatory ☐ Desirable
NFR-S4	The public tracking endpoint SHALL return only: reference number, current status, order date, basic summary. Internal IDs, staff information, and payment details SHALL NOT be exposed.	☑ Mandatory ☐ Desirable

NFR-S5	Passwords SHALL be stored as one-way hashes; JWT secret and database credentials SHALL be configurable via environment variables and SHALL NOT be committed to version control.	☑ Mandatory ☐ Desirable
NFR-S6	In production, the system SHALL use a strong JWT secret (≥32 characters) and a non-default database password.	☑ Mandatory ☐ Desirable

3.7.2 Performance

Table 15. Performance

ID	**Requirement**	**Priority**
NFR-P1	The system SHALL support concurrent use by at least two users (Admin and Staff) without degradation.	☑ Mandatory ☐ Desirable
NFR-P2	Order list and payment list SHALL support pagination to handle growth of data.	☑ Mandatory ☐ Desirable
NFR-P3	API responses SHALL complete within a reasonable time for single-shop workload (no formal SLA for MVP).	☐ Mandatory ☑ Desirable

3.7.3 Availability & Recovery

Table 16. Availability & Recovery

ID	Requirement	Priority
NFR-A1	The system SHALL be deployable and operable on hardware available to the client (single machine or local network).	☑ Mandatory ☐ Desirable
NFR-A2	Database backup SHALL be supported via documented script; backup format SHALL allow restore to PostgreSQL.	☑ Mandatory ☐ Desirable
NFR-A3	Deployment and restore procedures SHALL be documented in the Deployment Guide.	☑ Mandatory ☐ Desirable

3.7.4 Audit & Traceability

Table 17. Audit & Traceability

ID	Requirement	Priority
NFR-T1	Every order status change SHALL be recorded with timestamp and user (audit trail in order status logs).	☑ Mandatory ☐ Desirable
NFR-T2	Payment records SHALL be linked to order and user who recorded the payment.	☑ Mandatory ☐ Desirable
NFR-T3	Reports SHALL be computed from recorded payment data only (no estimates or manual overrides in MVP).	☑ Mandatory ☐ Desirable

3.7.5 Usability

Table 18. Usability

ID	Requirement	Priority
NFR-U1	Admin and Staff SHALL be able to perform core tasks (create order, update status, record payment, view reports) with minimal training.	☑ Mandatory ☐ Desirable
NFR-U2	An end-user manual SHALL be provided.	☑ Mandatory ☐ Desirable
NFR-U3	Error messages SHALL be clear enough for staff to correct invalid input (e.g., payment amount mismatch, invalid status transition).	☐ Mandatory ☑ Desirable

3.7.6 Maintainability & Operations

Table 19. Maintainability & Operations

ID	Requirement	Priority
NFR-M1	System architecture, API contract (OpenAPI), and data model (ERD) SHALL be documented and treated as source of truth.	☑ Mandatory ☐ Desirable
NFR-M2	Database schema changes SHALL be version-controlled via Flyway migrations.	☑ Mandatory ☐ Desirable
NFR-M3	The system SHALL support configuration via environment variables without code changes.	☑ Mandatory ☐ Desirable

NFR-M4	A handover checklist and deployment guide SHALL be provided for production deployment and client handover.	☑ Mandatory ☐ Desirable

Chapter 4: System Design

4.1 Technology Stack

This document outlines the software, frameworks, and tools used to develop the Faith Laundry Shop Management System, along with the justification for their selection.

4.1.1 Frontend Development

- Framework: Next.js (React)
- Styling: Tailwind CSS
- Language: TypeScript
- Justification: Next.js provides a robust, fast, and scalable framework for building modern web applications. TypeScript ensures type safety, reducing runtime errors. Tailwind CSS allows for rapid UI development and prototyping.

4.1.2 Backend Development

- Framework: Spring Boot 3.5
- Language: Java 21
- Architecture: RESTful API (MVC Pattern)
- Justification: Java 21 and Spring Boot offer an enterprise-grade, secure, and highly reliable backend environment. Spring Data JPA simplifies database interactions, and Spring Security provides robust authentication mechanisms.

4.1.3 Database

- Database Management System: PostgreSQL
- Migration Tool: Flyway

- Justification: PostgreSQL is a powerful, open-source object-relational database system known for reliability, feature robustness, and performance. Flyway ensures database schemas remain consistent across environments.

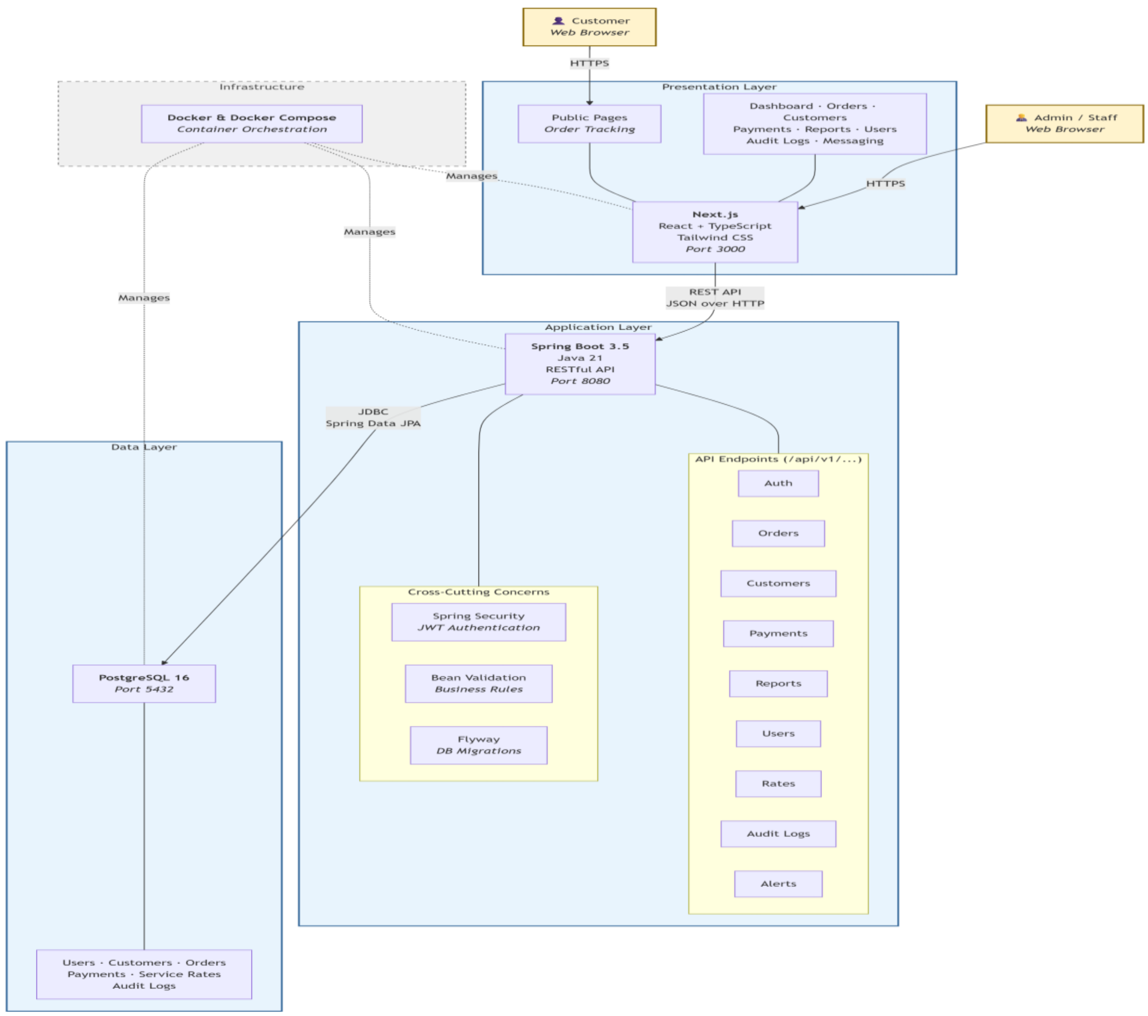
4.1.4 Development & Deployment Tools

- Version Control: Git & GitHub
- Containerization: Docker & Docker Compose
- Testing: JUnit 5, Testcontainers (Backend), Jest (Frontend)
- Justification: Docker ensures that the system runs identically in development, testing, and production environments, eliminating the "it works on my machine" problem.

4.2 System Architecture

This describes the high-level architecture of the Faith Laundry Shop Management System.

Figure 4. System Architecture Diagram



4.2.1 Architectural Pattern

The system follows a standard Client-Server Architecture utilizing a modern API-driven approach.

4.2.1.1 Presentation Layer (Frontend)

- Built with Next.js.
- Runs in the browser of the Admin or Staff.

- Communicates with the backend exclusively via RESTful JSON APIs.
- Responsible for routing, UI rendering, form validation, and user interaction.

4.2.1.2 Application Layer (Backend)

- Built with Spring Boot.
- Acts as the central hub for business logic, authentication, and database transactions.
- Exposes secure endpoints (/api/v1/...).
- Validates all incoming data ensuring business rules (like the 8kg per load limit) are strictly enforced at the server level.

4.2.1.3 Data Layer (Database)

- Uses PostgreSQL.
- Stores persistent data including user credentials, customer profiles, laundry orders, and payment histories.
- Data integrity is maintained via foreign keys, constraints, and transactions.

4.2.2 Deployment Architecture

The system is containerized using Docker, allowing the frontend, backend, and database to run as isolated but interconnected services on the host machine or cloud provider.

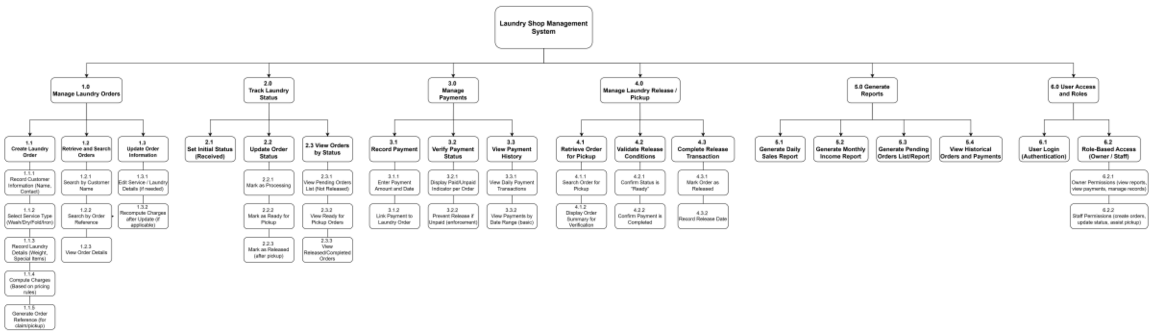
4.3 Functional Decomposition Diagram

4.3.1 Overview

The Functional Decomposition Diagram (FDD) provides a top-down view of the Laundry Shop Management System's functions. It visually breaks down the overall system into smaller, manageable subsystems such as Order Management, Customer Management, Payment Processing, Sales & Reporting, and User Management.

4.3.2 Diagram

Figure 5. Functional Decomposition Diagram



4.4 Data Flow Diagram

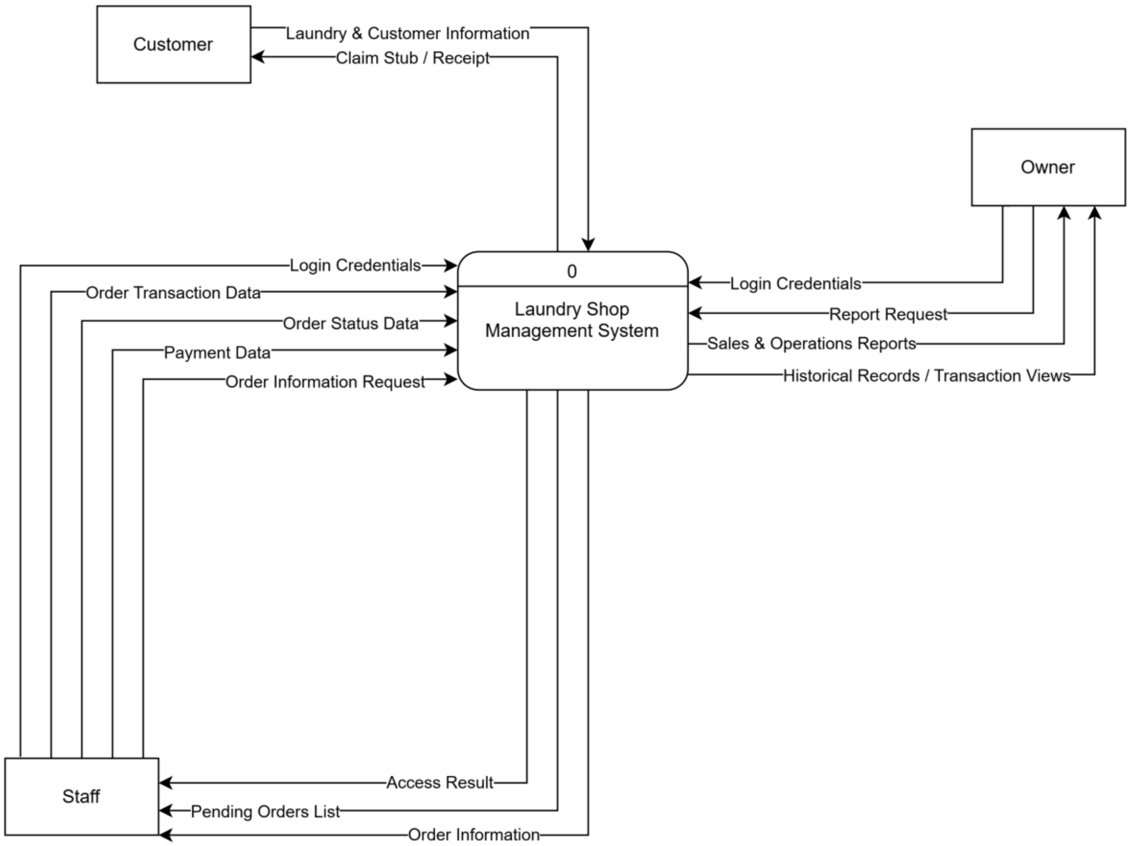
4.4.1 Overview

The Data Flow Diagrams (DFD) visually represent the flow of information through the Laundry Shop Management System. It models external entities (Admin, Staff, Customer), the processes that transform data, and the data stores where information is kept.

4.4.2 Context Diagram (DFD Level 0)

The Level 0 DFD, or Context Diagram, models the entire system as a single process and demonstrates its interactions with external entities.

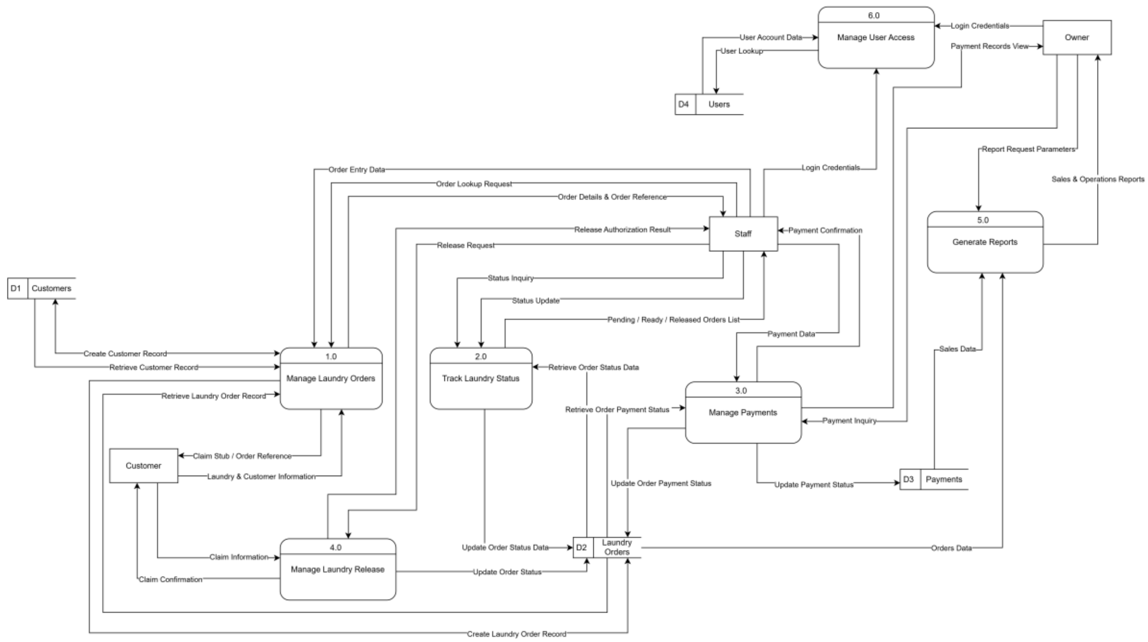
Figure 6. Context Diagram / DFD Level 0



4.4.3 Level 1 Data Flow Diagram (DFD Level 1)

The Level 1 DFD decomposes the main system process into distinct sub-processes (e.g., Order Processing, Payment Processing, Reporting) to provide greater detail about data movement and storage.

Figure 7. DFD Level 1



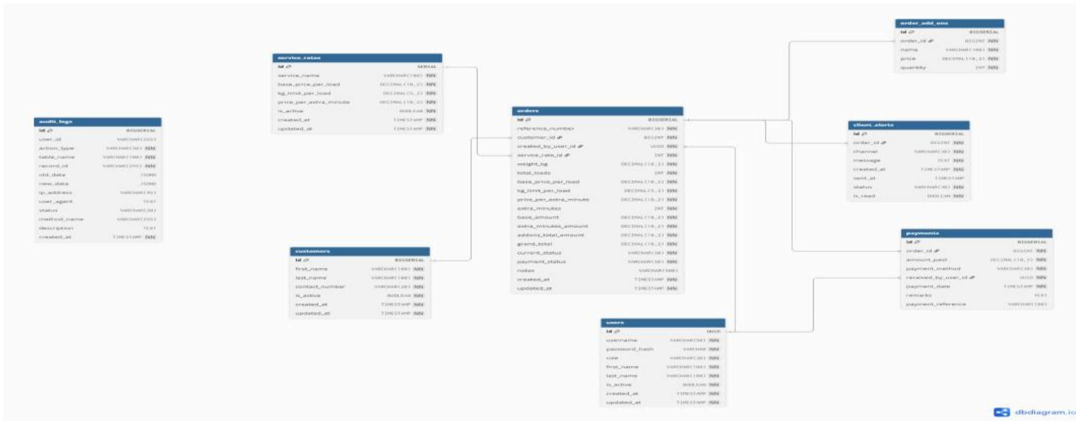
4.5 Entity Relationship Diagram

4.5.1 Overview

The Entity Relationship Diagram (ERD) illustrates the logical and physical data model supporting the Laundry Shop Management System. It defines the relationships between primary entities such as Users, Laundry Orders, Customers, and Payments.

4.5.2 Diagram

Figure 8. Entity Relationship Diagram



4.6 User Interface Design

4.6.1 Design Philosophy

The user interface of the Faith Laundry Shop Management System was designed following established Human-Computer Interaction (HCI) principles to ensure usability, learnability, and visual clarity for non-technical shop staff. The design adopts a Command Center paradigm where the dashboard serves as the central operational hub, providing at-a-glance visibility into the entire laundry workflow.

Key HCI principles applied throughout the interface include Nielsen's Heuristics for visibility of system status, match between the system and the real world, and aesthetic-minimalist design. The interface uses an F-pattern visual hierarchy optimized for rapid scanning during peak shop hours.

4.6.2 Design System

4.6.2.1 Color Palette

The visual identity of the system is anchored to a purpose-driven color palette. Each color is mapped to a specific semantic meaning to reduce cognitive load and support rapid decision-making.

Table 20. Color Palette

Token	Hex Value	Usage
Brand Blue	#15489D	Primary actions, navigation, branding
Brand Cyan	#30A8D4	Secondary actions, interactive highlights
Success Green	#047857	Ready status, completed actions, positive KPIs
Warning Amber	#B45309	Pending states, caution indicators
Error Rose	#BE123C	Error states, destructive actions
Neutral 50	#F8FAFC	Page backgrounds
Neutral 900	#0F172A	Primary text

4.6.2.2 Typography

The system uses a dual-font strategy for visual hierarchy. Display headings use a geometric sans-serif typeface for bold, attention-grabbing titles, while body text uses Inter for maximum legibility at small sizes. All text sizes follow an 8-point modular scale to maintain consistent vertical rhythm across screens.

4.6.2.3 Visual Effects

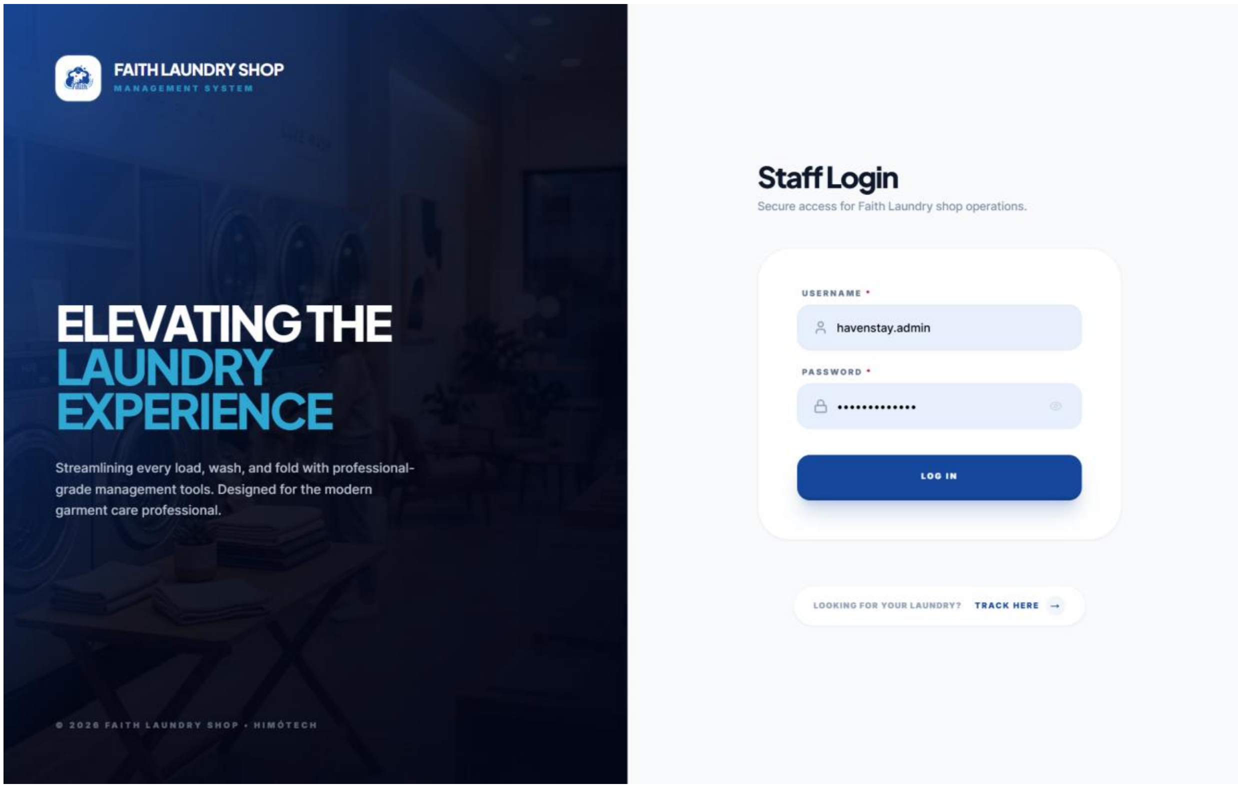
The interface employs glassmorphism as a unifying design language. Glass panels use a combination of semi-transparent white backgrounds, backdrop blur filters, and subtle inner borders to create depth without visual clutter. This technique is applied consistently to cards, modals, and the sidebar navigation.

4.6.3 Screen Layouts

4.6.3.1 Login Page

The Login Page serves as the staff authentication portal. It follows a vertically centered, single-column layout that minimizes visual distractions and directs focus entirely to the credential input fields. The design applies Nielsen's Aesthetic-Minimalist heuristic by removing all unnecessary elements. Error messages are displayed inline using a high-contrast rose alert banner with an icon for immediate visibility. A secondary call-to-action below the form links customers to the public order tracking page.

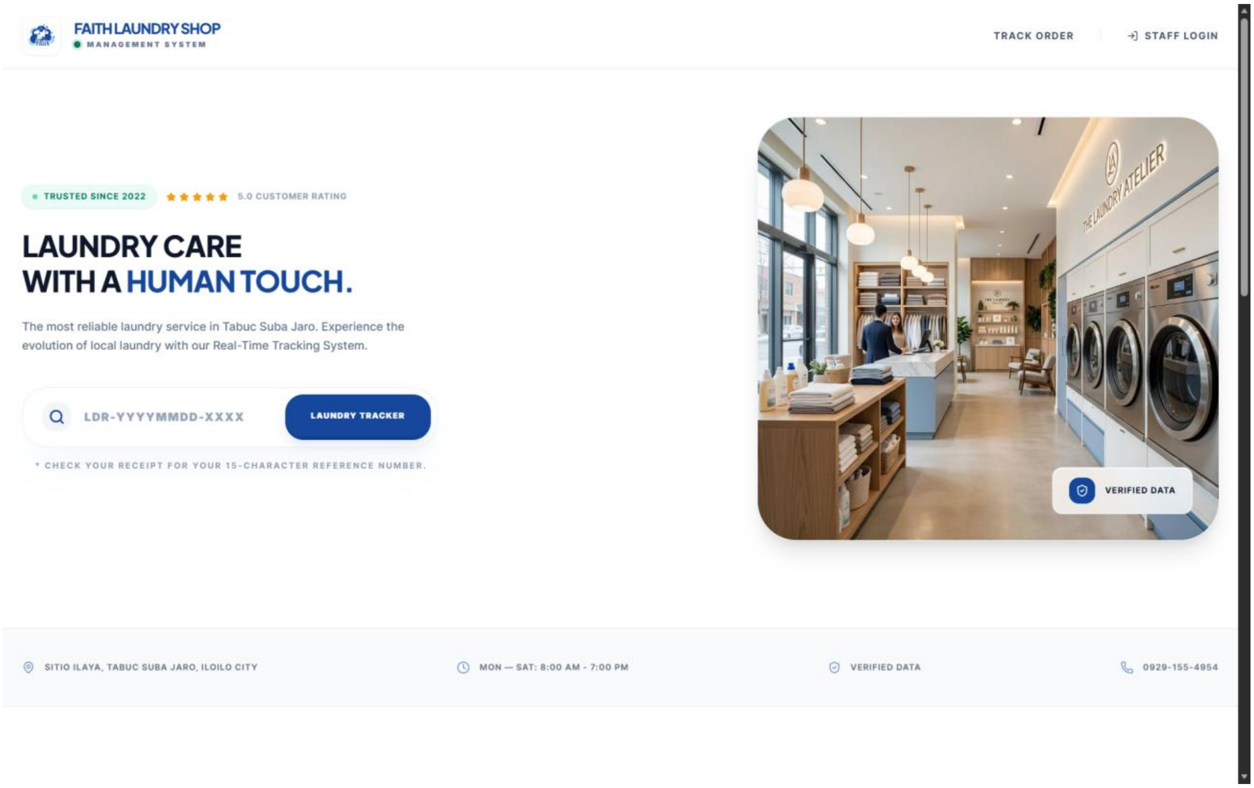
Figure 9. Login Page



4.6.3.2 Public Landing Page

The Public Landing Page is the customer-facing entry point to the system. It features a full-width hero section with the business name, service description, and an integrated order tracking search bar. Below the hero, a location strip displays the shop address, operating hours, and contact information. A features section highlights the value propositions of the service, and a pricing card dynamically displays the current standard wash rate fetched from the backend API. The page concludes with a footer containing business location details and support contact information.

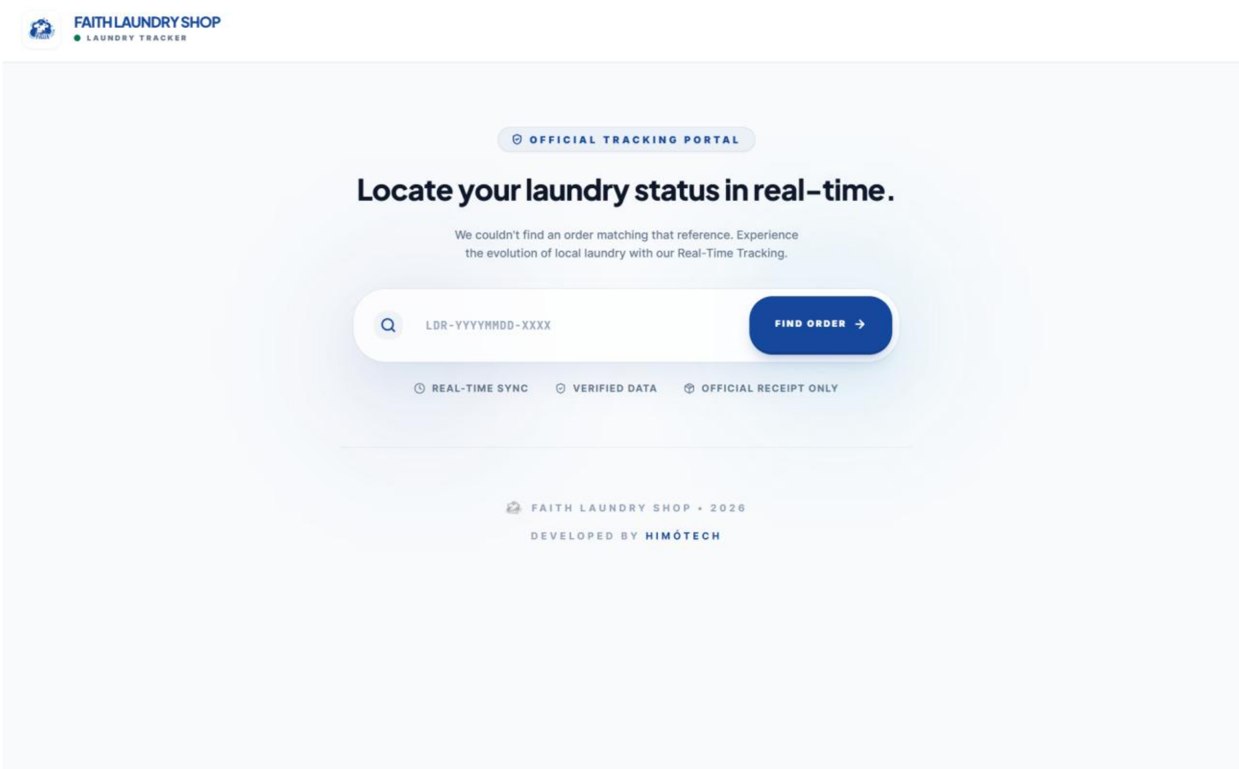
Figure 10. Landing Page



4.6.3.3 Order Tracking Page

The Order Tracking Page allows customers to look up the real-time status of their laundry order using a reference number. Upon entering a valid reference number, the page displays the current order status using a visual progress stepper, the service type, weight, pricing breakdown, and a timestamped status history. This page is publicly accessible and does not require authentication.

Figure 11. Order Tracking Page



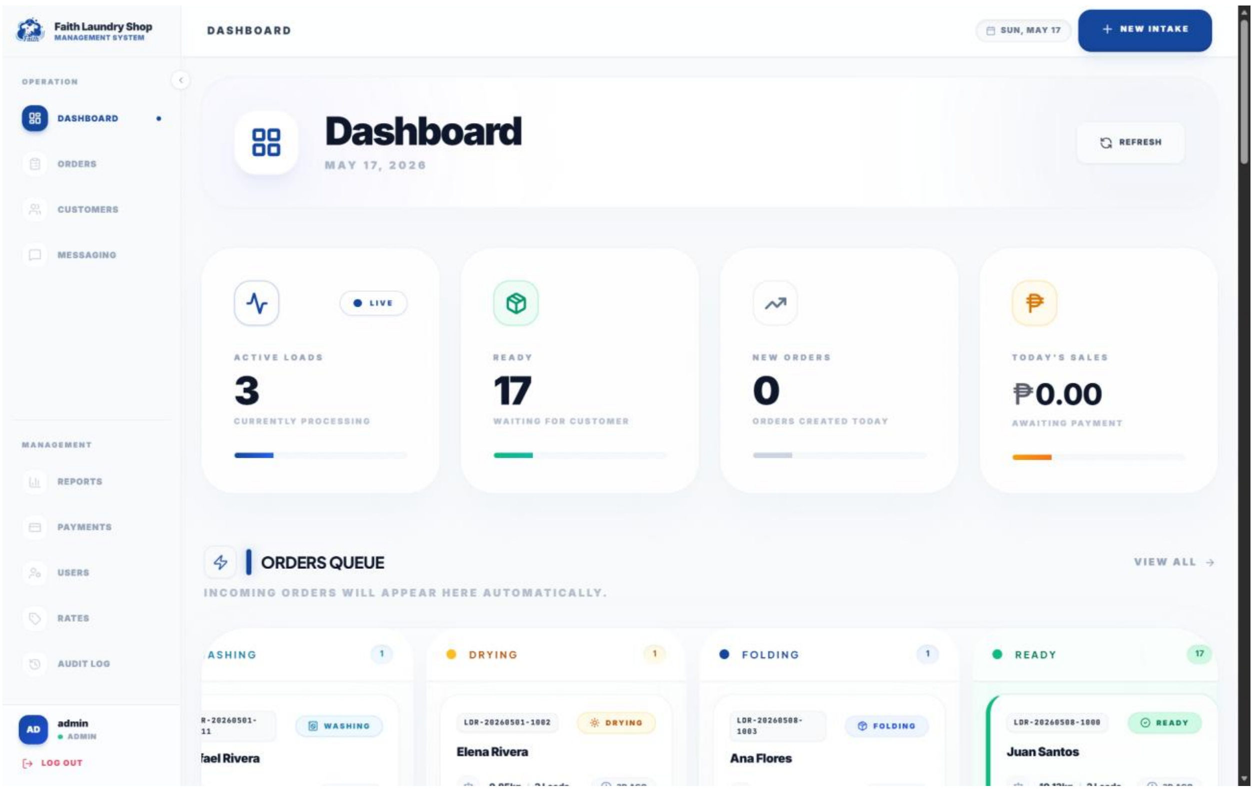
4.6.3.4 Dashboard

The Dashboard is the primary operational interface for staff and administrators. It is composed of two main sections. The top section displays Key Performance Indicator (KPI) cards showing the count of active loads, orders ready for pickup, today's new orders, and today's revenue (visible only to administrators). These cards use color-coded variants to communicate status at a glance.

The bottom section features the Order Pipeline, a five-column Kanban-style board that visualizes all active orders organized by their lifecycle status: Received, In Progress, Ready for

Pickup, Released, and Cancelled. Staff can advance orders through the pipeline using action buttons on each order card, triggering backend status transitions and audit log entries.

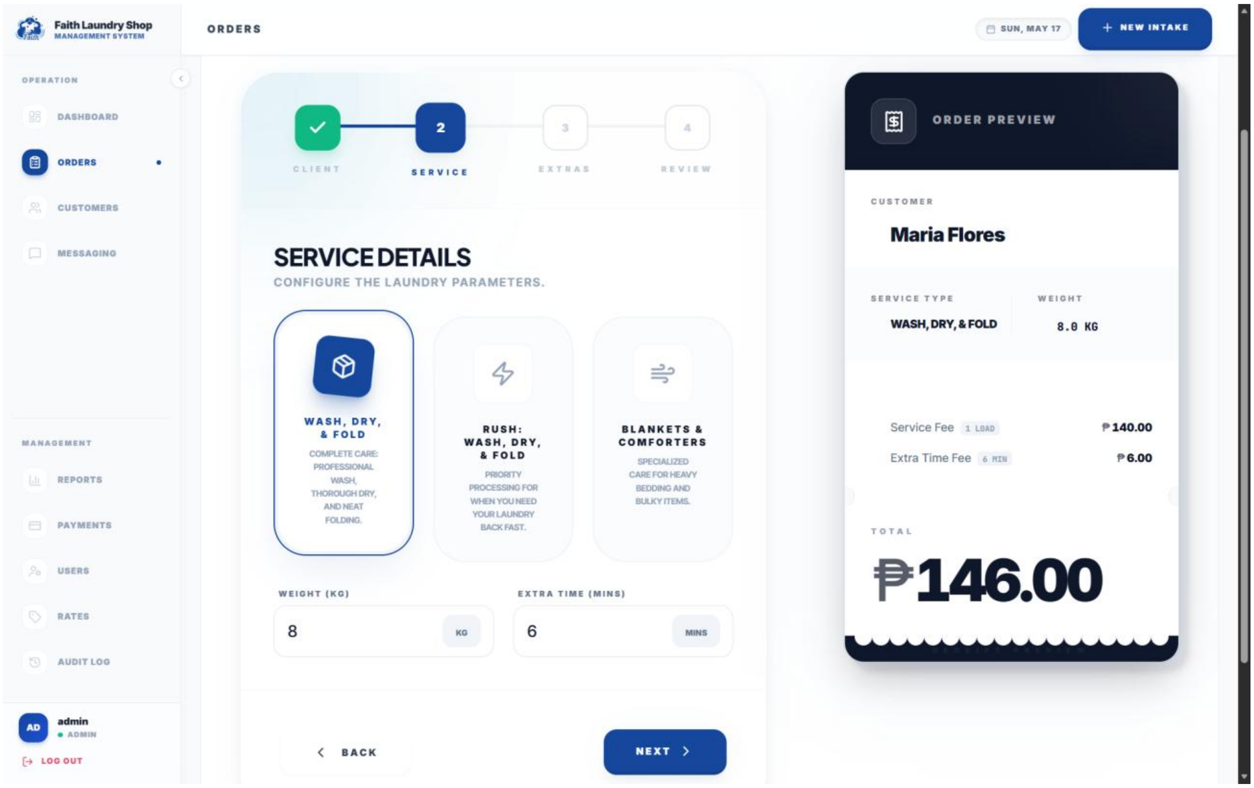
Figure 12. Dashboard



4.6.3.5 Order Intake Wizard

The Order Intake Wizard provides a guided, multi-step form for creating new laundry orders. The wizard is divided into sequential steps: customer selection or creation, service and weight input, optional add-ons, and a final review and confirmation step. Each step validates input before allowing progression to the next, preventing incomplete order submissions. The wizard dynamically calculates pricing based on the selected service rate, weight, and any add-ons.

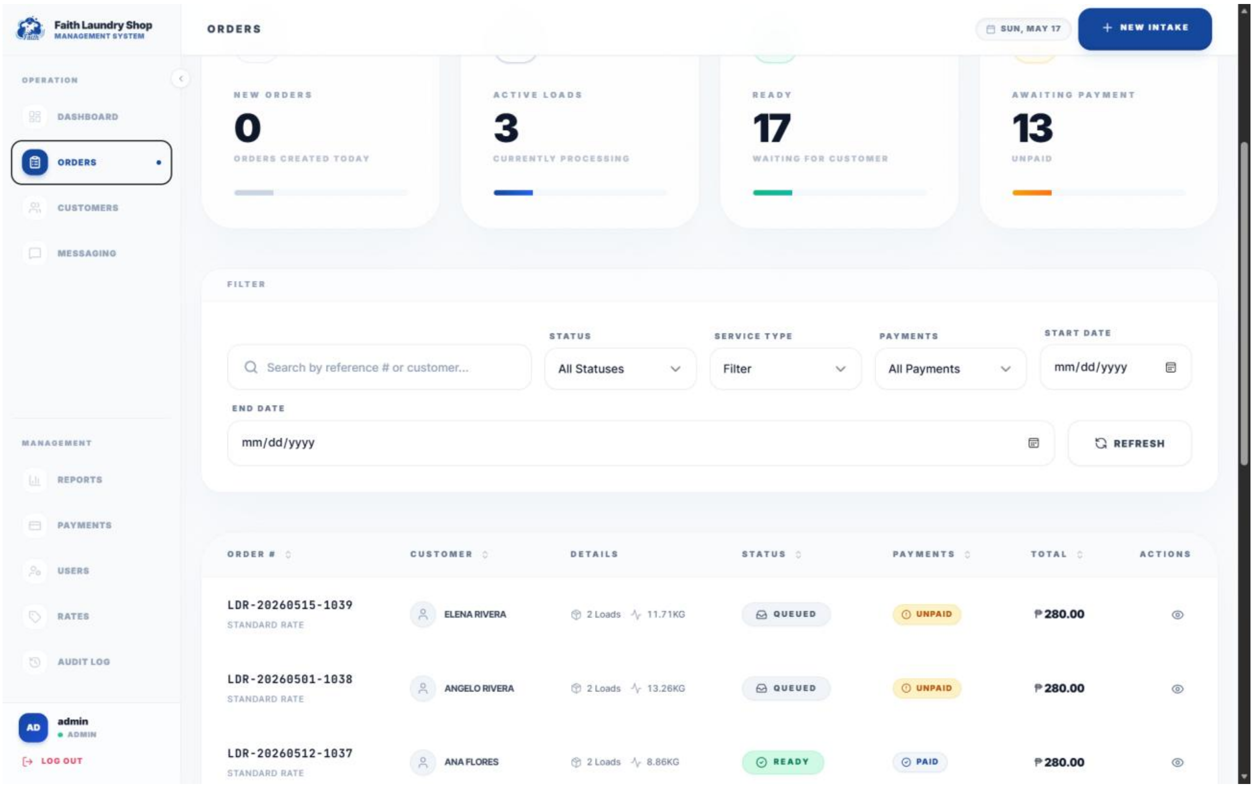
Figure 13. Order Intake Wizard



4.6.3.6 Order Management

The Order Management displays all orders in a searchable, filterable, and paginated table. Each row shows the reference number, customer name, service type, total amount, current status, and payment status. Status badges use semantic color coding consistent with the design system. Staff can click on any order to view its full details, update its status, or process a payment.

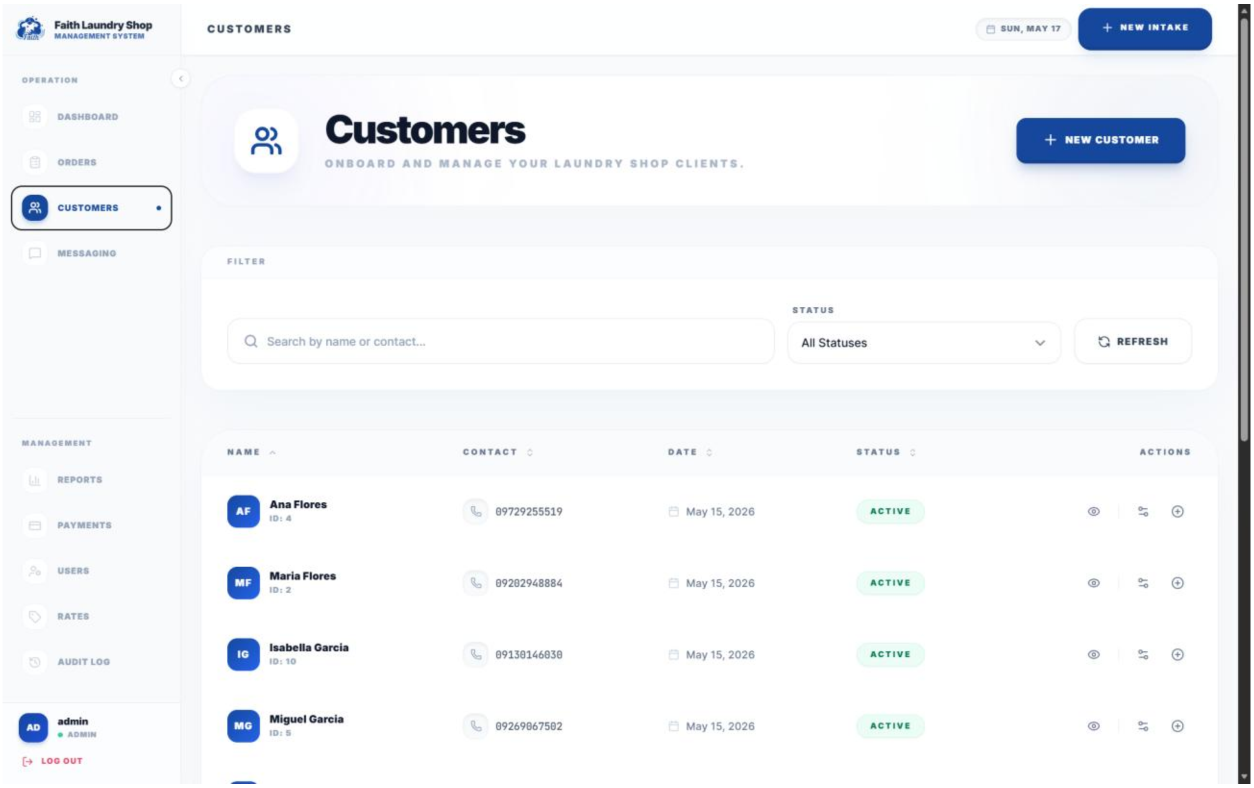
Figure 14. Order Management



4.6.3.7 Customer Management

The Customer Management module provides a master list of all registered customers with their contact information and order history. Staff can search customers by name or contact number, view individual customer profiles, and edit customer details. The customer detail view displays the complete order history for that customer along with summary statistics.

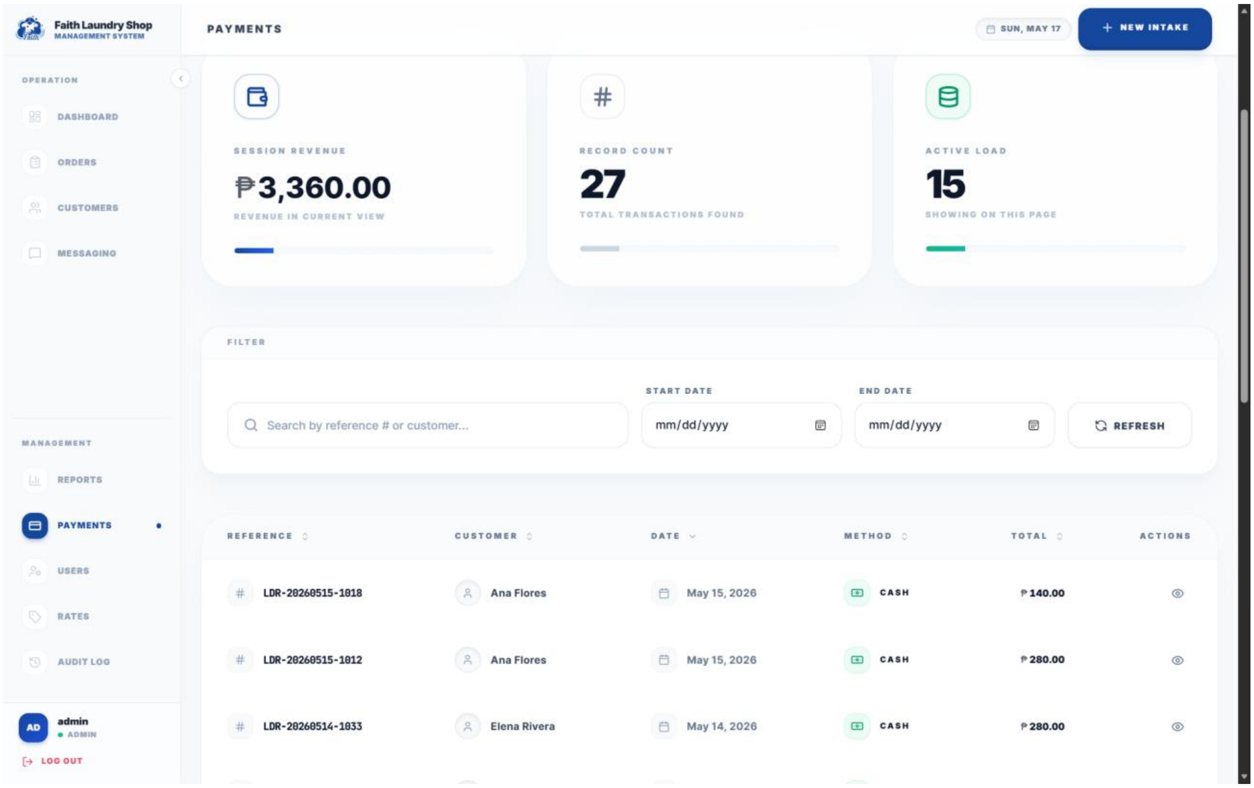
Figure 15. Customer Management



4.6.3.8 Payment Management

The Payment Management module displays a complete ledger of all payment transactions. The table includes the order reference number, customer name, amount paid, payment method, the staff member who received the payment, and the payment date. A payment action modal allows staff to record payments for unpaid orders, capturing the payment method and optional remarks.

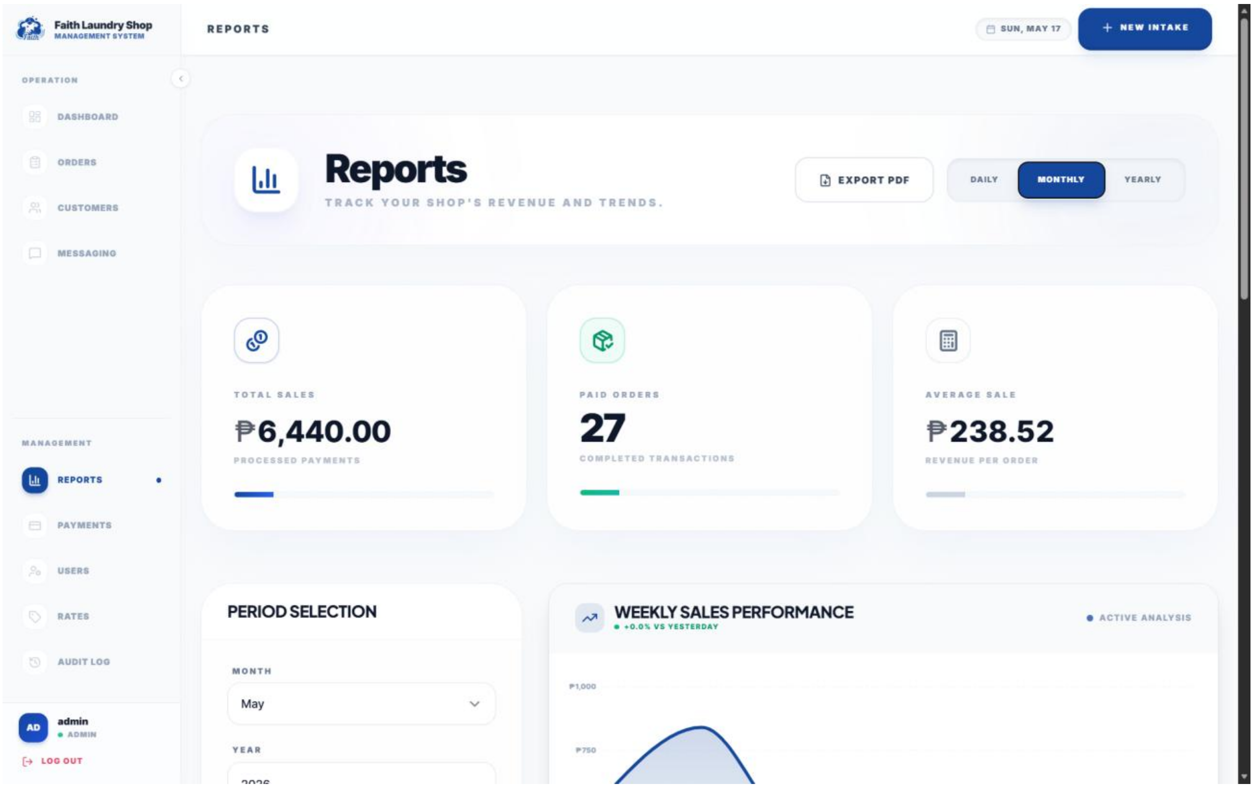
Figure 16. Payment Management



4.6.3.9 Reports Module

The Reports Module provides automated sales reporting capabilities. It displays revenue charts, order volume trends, and service type breakdowns across configurable date ranges. The module supports daily and monthly aggregation views with exportable data. Revenue figures, order counts, and average order values are displayed as KPI summary cards above the chart area. This module is restricted to administrators only.

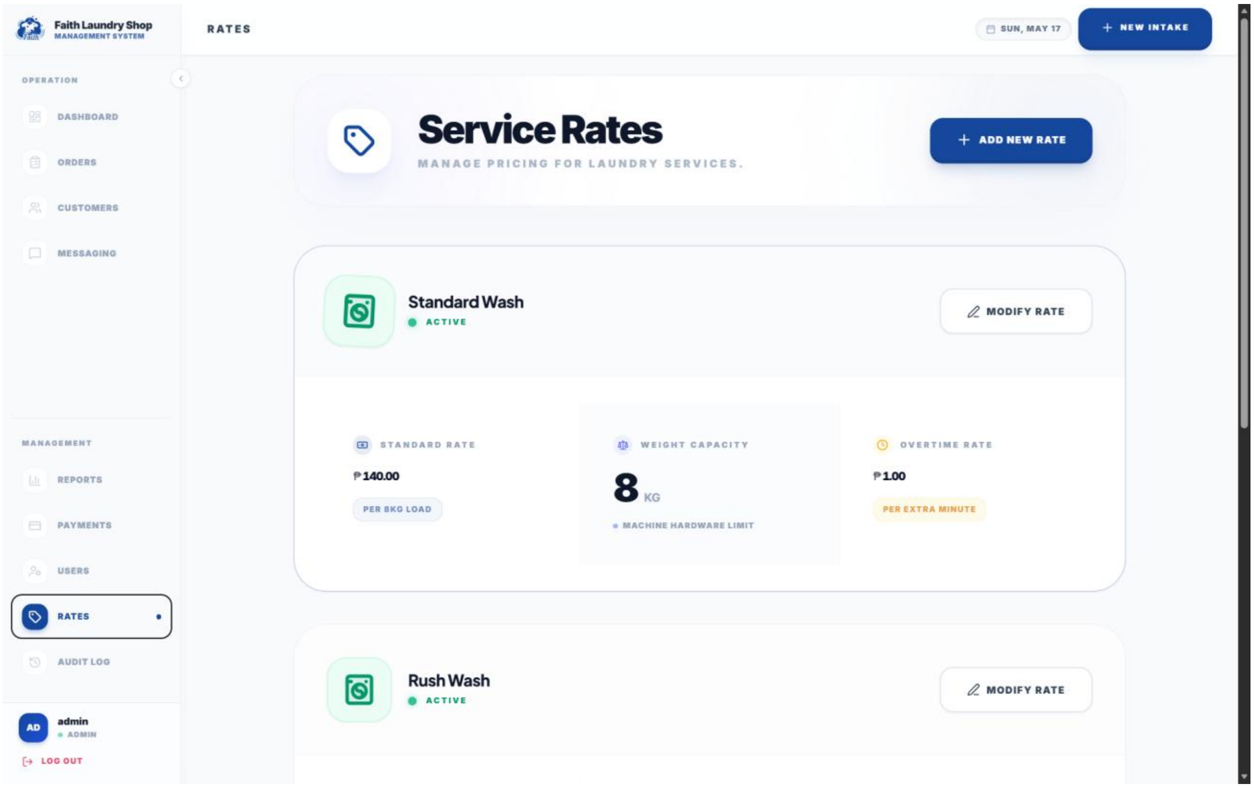
Figure 17. Reports Module



4.6.3.10 Service Rates Configuration

The Service Rates Configuration screen allows administrators to manage the pricing structure of the laundry shop. Each service rate entry defines the service name, base price per load, weight limit per load, and extra time charge per minute. Administrators can add new service types, edit existing rates, and activate or deactivate services. Changes to service rates are logged in the audit trail for accountability.

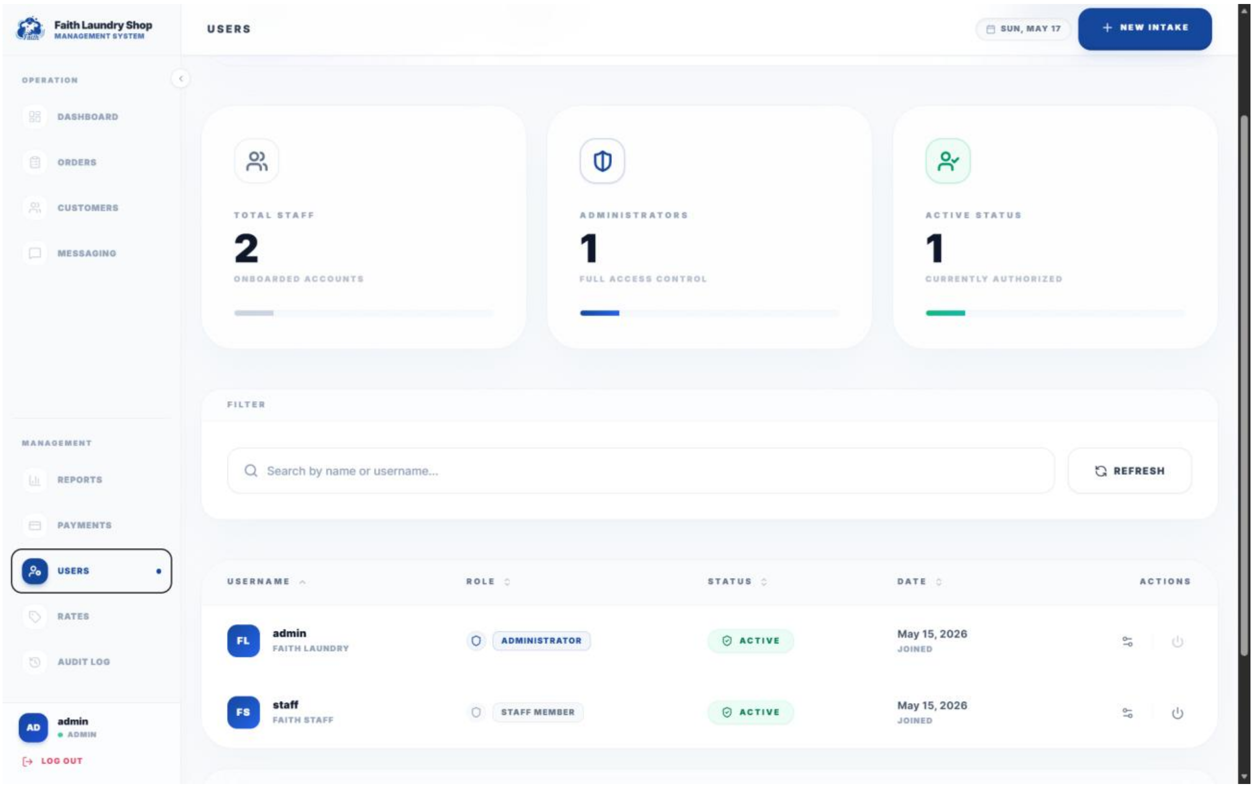
Figure 18. Service Rates Configuration



4.6.3.11 User Management

The User Management module allows administrators to manage staff accounts. The interface displays all system users with their usernames, full names, roles, and active status. Administrators can create new user accounts, edit existing user profiles, reset passwords, and deactivate accounts. Role-based access control is enforced at both the frontend and backend levels.

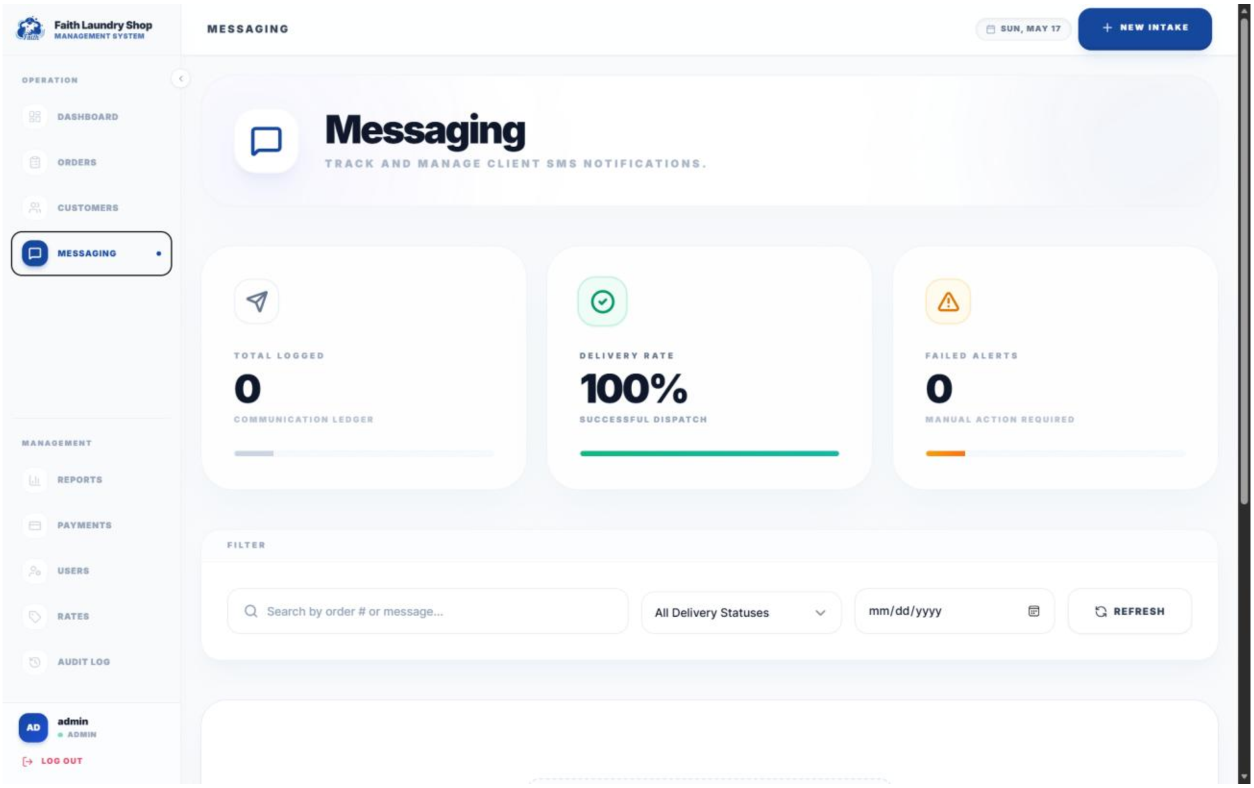
Figure 19. User Management



4.6.3.12 Client Alerts (Messaging)

The Client Alerts module provides a log of all customer notifications generated by the system. Each alert entry shows the associated order reference, the notification channel, message content, creation timestamp, and delivery status. This module enables staff to monitor customer communication history and verify that status update notifications have been sent successfully.

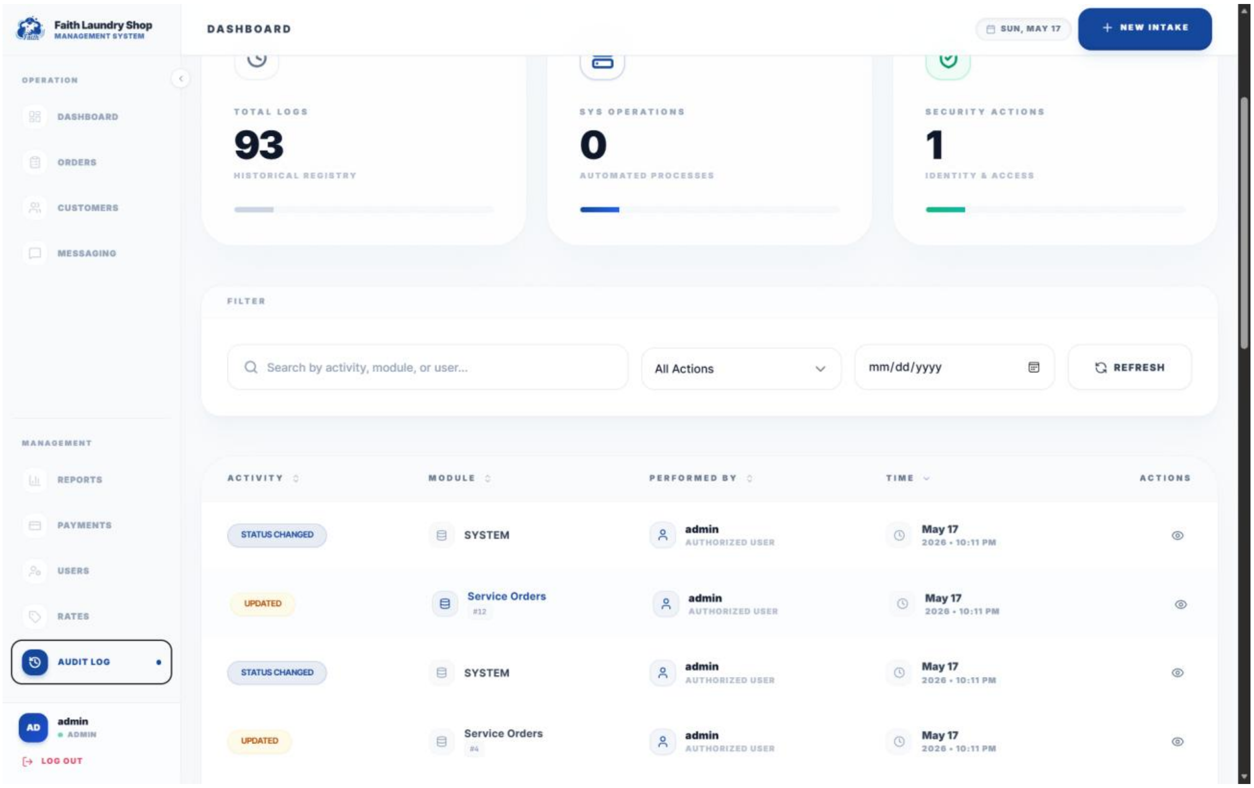
Figure 20. Client Alerts



4.6.3.13 Audit Logs

The Audit Logs module provides a forensic trail of all system actions for accountability and compliance. The log displays the user who performed the action, the action type, the affected table and record, timestamps, and before-and-after data snapshots in JSON format. This module is restricted to administrators and supports filtering by action type, table name, and date range.

Figure 21. Audit Logs



4.6.4 Responsive Design

The interface is fully responsive across desktop, tablet, and mobile viewports. On desktop screens, the layout uses a persistent sidebar navigation with a collapsible state. On tablet and mobile viewports, the sidebar transitions to a slide-out drawer accessible via a hamburger menu icon in the mobile top navigation bar. All data tables, forms, and card grids gracefully adapt their column counts and spacing to fit smaller screens without loss of functionality.

4.6.5 Accessibility Considerations

The system implements several accessibility features aligned with WCAG 2.1 Level AA guidelines. All interactive elements have visible focus indicators using consistent focus ring styles. Form inputs include descriptive labels and error messages. Color is never used as the sole indicator of status; text labels and icons always accompany color-coded badges. The interface maintains a minimum contrast ratio of 4.5:1 for all text content against its background

Chapter 5: Testing

5.1 Test Plan and Test Cases

5.1.1 Testing Strategy

To ensure the Faith Laundry Shop Management System is reliable and bug-free, a multi-tiered testing strategy is employed:

- Unit Testing: Tests individual functions, classes, and services in isolation (e.g., ensuring the load calculation logic correctly computes 9kg as 2 loads).
- Integration Testing: Tests how different parts of the system work together (e.g., verifying that creating a new order successfully saves to the PostgreSQL database).
- User Acceptance Testing (UAT): The final phase where the end-users (Admin and Staff) test the system in a real-world scenario to ensure it meets their business needs.

5.1.2 Sample Test Cases

5.1.2.1 TC-01: User Authentication

- Objective: Verify that only authorized staff can access the system.
- Steps:
  1. Navigate to the Login page.
  2. Enter a valid username and password.
  3. Click "Sign In."

- Expected Result: System authenticates the user and redirects to the Dashboard (Overview) page. A JWT token is stored in the session.

5.1.2.2 TC-02: Order Creation (Standard Load)

- Objective: Verify that an order within the 8kg limit computes as 1 load at the correct rate.
- Steps:
  1. Login as Staff.
  2. Navigate to "New Order."
  3. Select or create a customer.
  4. Choose "Standard Wash" service and set weight to 7kg.
  5. Submit the order.
- Expected Result: System registers 1 load and sets the base amount to ₱140.00. The order status is set to RECEIVED and a reference number in the format LDR-XXXXXXXX-XXXX is generated.

5.1.2.3 TC-03: Order Creation (Excess Load)

- Objective: Verify that an order exceeding 8kg computes additional loads correctly.
- Steps:
  1. Login as Staff.
  2. Navigate to "New Order."
  3. Select or create a customer.

4. Choose "Standard Wash" service and set weight to 9kg.
5. Submit the order.

- Expected Result: System registers 2 loads and sets the base amount to ₱280.00 (2 loads × ₱140.00/load).

5.1.2.4 TC-04: Order Status Advancement

- Objective: Verify that orders can be advanced through the pipeline in the correct sequence.
- Steps:

1. Login as Staff.
2. Navigate to the Dashboard.
3. Locate a RECEIVED order in the pipeline.
4. Click the advance action button on the order card.

- Expected Result: The order moves from RECEIVED to IN_PROGRESS. The status change is recorded in the audit log with the user ID and timestamp.

5.1.2.5 TC-05: Payment Before Release

- Objective: Ensure an order cannot be released without recording payment first.
- Steps:

1. Locate a READYFORPICKUP order with payment status UNPAID.
2. Attempt to advance the order to RELEASED.

- Expected Result: The system intercepts the action and opens a payment modal requiring the staff to record the payment before the order can be released.

5.1.2.6 TC-06: Payment Recording

- Objective: Verify that a payment can be recorded with the correct amount and method.
- Steps:
  1. Open an unpaid order that is READYFORPICKUP.
  2. Click the release action, which opens the payment modal.
  3. Select "CASH" as payment method.
  4. Confirm the payment.
- Expected Result: Payment is recorded in the payments table. The order's payment status changes to PAID, and the order advances to RELEASED.

5.1.2.7 TC-07: Customer Duplicate Prevention

- Objective: Verify that duplicate customer records cannot be created.
- Steps:
  1. Login as Staff.
  2. Navigate to "New Order" and create a customer with name "Juan Dela Cruz" and contact "09171234567."
  3. Attempt to create another customer with the same name and contact number.
- Expected Result: System prevents the duplicate entry due to the unique constraint on (lastname, firstname, contact_number) and returns an appropriate error message.

5.1.2.8 TC-08: Admin-Only Access Control

- Objective: Verify that Staff users cannot access Admin-restricted modules.
- Steps:
  1. Login as Staff.
  2. Attempt to navigate to the Reports page.
  3. Attempt to navigate to the User Management page.
- Expected Result: The system either hides these navigation items from the Staff user or redirects with an unauthorized error if accessed directly via URL.

5.1.2.9 TC-09: Audit Log Generation

- Objective: Verify that all order and payment changes are recorded in the audit trail.
- Steps:
  1. Login as Admin.
  2. Create a new order.
  3. Navigate to the Audit Logs module.
  4. Search for the newly created order's record ID.
- Expected Result: The audit log contains an INSERT entry for the orders table with the full order data captured in the new_data JSON field, including the user who performed the action.

5.1.2.10 TC-10: Public Order Tracking

- Objective: Verify that customers can track their order using the reference number without logging in.
- Steps:
  1. Open the public landing page (without authentication).
  2. Enter a valid order reference number (e.g., LDR-20260517-0001) in the tracking search bar.
  3. Submit the search.
- Expected Result: The system displays the order's current status, service type, weight, pricing breakdown, and a timestamped status history. No sensitive internal data is exposed.

Chapter 6: Deployment And User Manual

6.1 User Manual

6.1.1 System Access

1. Open a modern web browser (Google Chrome or Microsoft Edge recommended).
2. Navigate to the provided local network address or domain (e.g., http://localhost:3000).
3. Enter your assigned Username and Password on the Login screen.
4. Click Sign In to access the Dashboard.

6.1.2 Staff Operations

6.1.2.1 Dashboard Overview

Upon login, the Dashboard (Command Center) is displayed. It provides:

- KPI Cards showing active loads in progress, orders ready for pickup, today's new orders, and today's revenue (Admin only).
- Order Pipeline — a five-column Kanban board displaying all active orders organized by their lifecycle status: Received, In Progress, Ready for Pickup, Released, and Cancelled.

6.1.2.2 Creating a New Order

1. Click on New Order in the sidebar navigation.
2. Customer: Search for an existing customer by name or contact number, or create a new customer profile.

3. Service & Weight: Select the service type (e.g., Standard Wash, Rush Wash, Blankets), then enter the weight in kilograms (kg). The system will automatically compute the number of loads based on the 8kg-per-load limit.
4. Add-ons (Optional): Select any optional add-ons (e.g., extra fabric conditioner). Each add-on will be itemized and added to the total.
5. Review & Confirm: Review the auto-computed total price including the base amount, any extra-time charges, and add-ons. Click Create Order to submit.
6. The system generates a unique reference number in the format LDR-XXXXXXXX-XXXX for the customer's claim stub.

6.1.2.3 Advancing Order Status

1. Navigate to the Dashboard to view the Order Pipeline.
2. Locate the order in the appropriate status column.
3. Click the action button on the order card to advance it to the next status:

- Received → In Progress: Click "Start Processing" when the laundry is loaded into the machine.
- In Progress → Ready for Pickup: Click "Mark Ready" when the laundry is washed, dried, and folded.
- Ready for Pickup → Released: Click "Release" — if payment is still pending, the system will prompt for payment first.

6.1.2.4 Recording Payment and Releasing Orders

1. When a customer arrives to pick up their laundry, locate the order in the Ready for Pickup column on the Dashboard, or navigate to the Orders list and filter by status.
2. Click the Release action on the order card. If the order is unpaid, a Payment Modal will appear.
3. In the Payment Modal:
   - The total amount due is displayed automatically.
   - Select the Payment Method: Cash, GCash, or Bank Transfer.
   - If using a digital payment method, enter the reference number or trace ID.
   - Click Settle Payment to confirm.
4. Once payment is recorded, the order status advances to Released and the payment status changes to Paid.

6.1.2.5 Managing Customers

1. Navigate to the Customers section from the sidebar.
2. View the complete customer list with names and contact numbers.
3. Click on a customer row to view their profile and order history.
4. To edit customer details, click the edit action on the customer detail page.

6.1.2.6 Viewing Orders

1. Navigate to the Orders section from the sidebar.
2. The orders table displays all orders with reference number, customer name, service type, total amount, current status, and payment status.

3. Use the search bar to find orders by reference number or customer name.
4. Click on any order row to view its full detail page, including service breakdown, add-ons, status history, and payment information.

6.1.2.7 Client Alerts

1. Navigate to the Messaging section from the sidebar.
2. View a log of all customer notifications generated by the system.
3. Each entry shows the associated order reference, notification channel, message content, and delivery status.

6.1.3 Admin Operations

6.1.3.1 Viewing Reports

1. Login with Admin credentials.
2. Navigate to the Reports section from the sidebar.
3. Select the reporting period (Daily or Monthly) to view total revenue, number of paid orders, and average order value.
4. Revenue charts and order volume trends are displayed visually with exportable data.

6.1.3.2 Managing Service Rates

1. Navigate to the Rates section from the sidebar.
2. View all active and inactive service rate configurations.
3. To edit a rate, click on the rate entry and modify the base price per load, weight limit, or extra-minute charge.
4. To add a new service type, click the add action and fill in the required fields.

5. To deactivate a service, toggle the active status. Deactivated services will not appear as options during order creation.

6.1.3.3 Managing Users

1. Navigate to the Users section from the sidebar.
2. View all system user accounts with their usernames, full names, roles (Admin or Staff), and active status.
3. To create a new user, click the add action and provide the username, password, full name, and role.
4. To deactivate a user, toggle the active status on the user detail page. Deactivated users cannot log in.

6.1.3.4 Viewing Audit Logs

1. Navigate to the Audit Logs section from the sidebar.
2. The audit trail displays all system actions including order creation, status changes, payment recordings, and configuration updates.
3. Each entry shows the user who performed the action, the action type, the affected record, and the timestamp.
4. Use filters to narrow down logs by action type, table name, or date range.

6.1.4 Public Features (No Login Required)

6.1.4.1 Order Tracking

1. Visit the public landing page of the application.
2. Enter the order reference number (from the claim stub) into the tracking search bar.

Click Track Order to view the current order status, service details, pricing breakdown, and a timestamped status history

6.2 Technical Setup Guide

This guide is intended for developers or IT personnel deploying the Faith Laundry Shop Management System on a new machine or server.

6.2.1 Prerequisites

Ensure the following software is installed on the host machine:

- Docker & Docker Compose: For running the database and all application containers.
- Java Development Kit (JDK) 21: Only if running the backend locally outside Docker.
- Node.js (v18+) & npm: Only if running the frontend locally outside Docker.
- Git: For cloning the project repository.

6.2.2 Environment Variables

Create a .env file in the root directory by copying the .env.example file:

cp .env.example .env

Edit the .env file and configure the following variables:

Database Configuration:

DB_HOST=localhost

DBNAME=laundrydb

DBUSER=laundryuser

DBPASSWORD=laundrypasswordchangeme

DB_PORT=5433

Backend Configuration:

SPRING_PORT=8080

SPRINGPROFILESACTIVE=dev

JWT_SECRET=dev-secret-change-in-production-minimum-32-characters-required

ALLOWED_ORIGIN=http://localhost:3000

ALLOWEDORIGINPATTERNS=http://localhost:3000,http://localhost:3001,https://*.ngrok-free.app

Dev Seed Accounts (dev profile only):

SEEDADMINUSERNAME=admin

SEEDADMINPASSWORDHASH=replacewithbcrypthash

SEEDSTAFFUSERNAME=staff

SEEDSTAFFPASSWORDHASH=replacewithbcrypthash

APPSEEDADMIN_PASSWORD=admin123

APPSEEDSTAFF_PASSWORD=staff123

Frontend Configuration:

FRONTEND_PORT=3000

NEXTPUBLICAPI_URL=http://backend:8080/api

SMS Notifications (Optional):

SEMAPHOREAPIKEY=yourapikey_here

SEMAPHORESENDERNAME=FaithLaundry

6.2.3 Running via Docker (Recommended)

1. Open a terminal in the root project directory.
2. Run the command: docker-compose up -d --build
3. Wait for the containers to initialize. The PostgreSQL database must pass its health check before the backend starts.
4. The Backend API will be available at http://localhost:8080.
5. The Frontend App will be available at http://localhost:3000 (or the port defined in FRONTEND_PORT).

6.2.4 Running Locally (Without Docker)

6.2.4.1 Database

Start the PostgreSQL database container only:

```
docker-compose up -d db
```

6.2.4.2 Backend

```
cd backend
./mvnw spring-boot:run -Dspring-boot.run.profiles=dev
```

The backend will start on port 8080 and automatically connect to the database.

6.2.4.3 Frontend

```
cd frontend
npm install
npm run dev
```

The frontend will start on port 3000 and connect to the backend API.

6.2.5 Database Migrations

The backend utilizes Flyway for database migrations. When the Spring Boot application starts, it will automatically connect to the PostgreSQL database and run any pending .sql migration scripts found in backend/src/main/resources/db/migration/. No manual database creation is required other than providing the blank schema via Docker.

The initial migration (V1init.sql) creates all core tables, audit infrastructure, triggers, and seed data for service rates. The second migration (V2seedusers.sql) creates development user accounts using Flyway placeholders — these are only populated when the SPRINGPROFILES_ACTIVE is set to dev and the corresponding seed environment variables are provided.

6.2.6 Production Deployment

For production deployment (e.g., on Render), set the following environment variables in the hosting dashboard:

- SPRINGPROFILESACTIVE=prod
- DBHOST, DBPORT, DBNAME, DBUSER, DB_PASSWORD — pointing to the production PostgreSQL instance (e.g., Neon).
- JWT_SECRET — a cryptographically random string of at least 32 characters.
- ALLOWED_ORIGIN — the production frontend URL (e.g., https://your-app.vercel.app).

Do not set the SEED_* variables in production to avoid creating default privileged accounts.

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